Canadian Community Health Survey (CCHS)
Annual component

User guide
2017 Microdata file

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WHAT'S NEW IN THE 2017 CANADIAN COMMUNITY HEALTH SURVEY?

Content

The following are some of the modifications that were made to Canadian Community Health Survey (CCHS) – Annual component questionnaire in 2017. For more details on the changes, please read Chapter 3:

- The modules **Sedentary behaviors (SBE), Alcohol use during maternal experience (MXA) and Smoking during maternal experience (MXS), Injuries (INJ), Washington Group – Disability measure (WDM) and Food security (FSC)** were asked as two-year theme modules in 2017.
- Questions on certain chronic conditions - Heart disease, Intestinal or stomach ulcers, Bowel disorders, Type of bowel disease, Urinary incontinence, Allergies, Source(s) of allergy, hay fever/nasal allergy (Theme 2), have been added to the **Chronic Conditions (CCC)** module in 2017 as a two-year theme content.
- The modules **Pap smear test (PAP), Mammography (MAM), Colorectal cancer testing (CCT), Fruit and vegetable consumption (FVC), Use of protective equipment (UPE)** became one-year theme content.

Module Changes

- In the Administration (ADM) and Income (INC) module, the personal income questions were removed; only household income will be collected for households with more than one person. A question to gain consent from single person households to share their tax data was added.

- Minor modifications were made to the cancer screening modules: Pap Test (PAP), Mammography (MAM).

- The Colorectal cancer screening (CCS) module was overhauled and the name was changed to Colorectal cancer testing (CCT).

- The Sedentary activities (SAC) module was redesigned and re-named Sedentary behaviors (SBE) module.

- The Fruit and vegetable consumption (FVC) module was modified slightly to split each initial question into 2 questions.

- The Contact with health professionals (CHP) and Exposure to second hand smoke (ETS) modules were removed from core content and offered as optional content to provinces and territories.

- Three derived variables (DVs) in the Physical activities for adults module (PAA) and one DV in the Physical activities for youth module (PAY) have been replaced. The variables PAADVATR, PAADVARC, and PAADVATH measured the average amount of active transportation, recreational, and other activities, per day that those activities took place. That is, if a respondent had 120 minutes of recreational activities during 4 days of the reporting week, the variable PAADVARC was calculated as 120 minutes /4 active days = 30 minutes per active day. Similarly in PAY, the variable PAYDVAVG measured the average number of minutes of all physical activity of youth, per active day. Each of these four DVs have been replaced by measures of average physical activity over the week, regardless of the number of days active. That is, the number of minutes is divided by 7 days. The new DVs are: PAADVTRA, PAADVRCA, PAADVOTA, and PAYDVAV7.

New modules

- The optional module Workplace injury (INW) has not been selected since the CCHS Redesign in 2015. It deals with injuries that occurred at a job other than the current main job. It determines the industry, occupation and the most important duties of the job when the injury occurred.

For more information on CCHS content, please see Appendix A for a selection of optional content by provinces and territories

Documentation

- Errata: Errors that are discovered in the CCHS data files and products after dissemination are communicated to users in a cumulative error log called “CCHS Errata". The errata provides details on:
  - the products affected (e.g. master or share files or PUMF);
  - the years affected;
  - suggested corrections (if possible); and
  - steps to carry out to apply the correction

Since the release of the 2015 CCHS, there have been 16 additional entries to the errata document. The subjects of these entries are:

- PAY_100 and PAY_105 erroneous note
- HMC variables incorrectly set to ‘not stated’

- DIA_055/DIA_060 incorrectly set to ‘not applicable
- Error in ALWDVLTR , ALWDVSTR
- Error in HS_fmt.sas (SAS format file) for Master and Share
- Error in PMHDVCLA
- Error in UPEDVSKB
- Error in EHG2DVR3 and EHG2DVR9
- Error in PAADVATR, PAADVARC, AND PAADVATH
- Error in SXBDVPRT, SXBDVSTI, and SXBDVTST
- Error in LOP_015 to LOP_090
- Error in MEXDVBM6
- Error in SDC_020A-K, and SDCDVCGT
- Error in GEN_025
- Error in FLU_025A-K
- Error in DHHDVAOS

For your convenience, the errata document has an index on the first page. It lists modules affected and the years affected. To see if a particular variable is affected, follow the hyperlinks on the associated errata item numbers.

An up to date copy of the CCHS Errata can be obtained by contacting Health Statistics Division at 613-951-1746 or statcan.hd-ds.statcan@canada.ca.

1. INTRODUCTION

The Canadian Community Health Survey (CCHS) is a cross-sectional survey that collects information related to health status, health care utilization and health determinants for the Canadian population. It surveys a large sample of respondents and is designed to provide reliable estimates at the health region level. The CCHS underwent a major redesign that started in 2012 and was implemented in collection in 2015. Details of this redesign are provided in Chapter 3.

The survey's objectives are as follows:

- support health surveillance programs by providing health data at the national, provincial and health region levels;
- provide a single data source for health research on small populations and rare characteristics;
- timely release of information easily accessible to a diverse community of users; and
- create a flexible survey instrument that includes a rapid response option to address emerging issues related to the health of the population.

The CCHS data is always collected from persons aged 12 and over living in private dwellings in over 100 health regions covering all provinces and territories. Excluded from the sampling frame are individuals living on Indian Reserves and on Crown Lands, institutional residents, full-time members of the Canadian Forces, youth aged 12 to 17 living in foster homes, and residents of certain remote regions. The CCHS covers approximately 98% of the Canadian population aged 12 and over.

The purpose of this document is to facilitate the manipulation of the CCHS microdata files and to describe the methodology used. The CCHS produces three types of microdata files: master files, share files and public use microdata files (PUMF). The characteristics of each of these files are presented in this guide. The PUMF is released after two years of data collection and contains two years of data. The next PUMF file (release date to be determined) will include the data collected for the years 2017 and 2018.

Any questions about the data sets or their use should be directed to:

Electronic Products Help Line:	1-800-949-9491
For custom tabulations or general data support:	
Client Custom Services, Health Statistics Division:	613-951-1746
E-mail:	statcan.HD-DS.statcan@canada.ca
For remote access support:	613-951-1746
E-mail:	statcan.cchssm-esccds.statcan@canada.ca

Fax: 613-951-0792

2. BACKGROUND

In 1991, the National Task Force on Health Information cited a number of issues and problems with the health information system. The members felt that data was fragmented; incomplete, could not be easily shared, was not being analysed to the fullest extent, and the results of research were not consistently reaching Canadians.[1]

In responding to these issues, the Canadian Institute for Health Information (CIHI), Statistics Canada and Health Canada joined forces to create a Health Information Roadmap in 1999. From this mandate, the Canadian Community Health Survey (CCHS) was conceived. The format, content and objectives of the CCHS evolved through extensive consultation with key experts and federal, provincial and community health region stakeholders to determine their data requirements.[2] The survey started data collection in 2000.

To meet many data requirements, the CCHS had a two-year data collection cycle. Up to 2007, the first year of the survey cycle, designated by ".1", was a general population health survey, designed to provide reliable estimates at the health region level. The second year of the survey cycle, designated by ".2", had a smaller sample and was designed to provide provincial level results focused on specific health topics.

Starting in 2007, the regional component of the CCHS program began collection on an ongoing basis. The x.1 cycles of the CCHS were renamed "the annual component" of the CCHS and are now collected every year. The full title of a given annual cycle is "The Canadian Community Health Survey – Annual component, 20XX" and the short title is simply "CCHS –20XX".

The focused content component of the survey have been designated by the name of the survey followed by the topic of the themes covered by each survey (e.g., “Canadian Community Health Survey on Healthy Aging” or “CCHS – Healthy Aging”).

In 2012, a major redesign began which changed the survey sampling and collection strategy. Survey content was qualitatively tested or retested to ensure validity and reliability in preparation for a new 8 year content plan beginning in 2015.

1. Health Canada; Statistics Canada. 1999. *Health Information Roadmap: Responding to Needs*. p. 3.
2. Canadian Institute for Health Information; Statistics Canada. 1999. *Health Information Roadmap: Beginning the Journey*. ISBN 1-895581-70-2. p. 19.

3. CCHS REDESIGN IN 2015

In 2012, the redesign began with extensive consultations across Canada with key experts and federal, provincial and health region stakeholders. The goal was to gather and input proposed changes and collect detailed information on the data requirements and products of the various partners.

Below are the main changes resulting from the CCHS redesign:

- Content changes were made to most modules, but mainly to the core and theme modules. Roughly 70% of the modules that existed before the redesign have seen some level of change. Those changes vary, and have involved either minor tweaks or major changes to concepts, vocabulary, or response categories. Some new modules were created to allow for emerging concepts or data gaps identified by data sharing partners such as Health Canada, the Public Health Agency of Canada and the provincial and territorial ministries of health.
- Variable conventions have changed in 2015. The naming convention used for variables in the 2015 CCHS use a maximum of eight characters. As before, positions 1 to 3 make up the module or questionnaire section name, while position 4 indicates the variable type (underscore, D, F, C or G). Survey elements in the questionnaire (e.g. E, C, D, Q, N) now follow a uniform convention that increases in increments of five (e.g. CCC_005, CCC_010, CCC_015).
- All the derived variables were reviewed and specifications were updated with the new variable names. Content and conceptual changes were also reviewed and new derived variables were created for new content modules. The annual data collection strategy changed for 2015. Prior to 2015, there were six two-month collection periods. Starting in 2015, survey collection was divided into four non-overlapping three-month periods.
- Starting in 2015, changes to data collection in the North resulted in approximately half of the communities in each territory visited annually. As a result, the data is only representative in the territories after two years.

4. CONTENT STRUCTURE OF THE CCHS

In addition to sociodemographic and administrative data, the CCHS includes four content components, each of which addresses a different need: the Core, Theme, Optional, and Rapid Response. Appendix A lists the modules included in the 2017-2018 questionnaire by component.

The average length of a CCHS interview is estimated between 40 and 45 minutes.

Table 4.1 Length of survey by component

CCHS component	Length
Core content	**19 minutes**
Theme content	**8 minutes**
Optional content	**8 minutes**
Rapid Response content (cost recovery basis)	**2 minutes**

4.1 Core content

The CCHS core content component includes questions asked of respondents in all provinces and territories (unless otherwise specified). These questions will remain relatively stable in the questionnaire for a period of about six years up until 2021.

4.2 Theme content

The theme content is asked of all survey respondents and comprises modules, or in some cases groups of questions within modules, that are related to a specific topic. There are two types of theme collected annually: two-year theme modules which are asked over a period of two years and one-year theme modules that are asked over a one year period. The one-year theme changes annually. Combined, the one-year and two-year theme content takes about 8 minutes of interview time. Themes are reintroduced in the survey every two, four or six years, if required. This component enables CCHS to better plan its content in the medium term.

4.3 Optional content

The optional content component is designed to give provinces and territories the opportunity to select content that addresses their own public health priorities. Each province and territory selects modules for up to 8 minutes of content. Survey participants will be asked the optional content modules that are selected for the province or territory in which they live. Optional content is selected two years at a time. It should be noted that, unlike the modules included in the core or

theme content, the resulting data from the optional content modules should not be generalized across Canada[3].

Appendix B presents the selection results of the optional content for 2017 and 2018 by province or territory of residence.

4.4 Rapid response content

The rapid response component is offered on a cost-recovery basis to organizations interested in obtaining national estimates on an emerging or specific topic related to the health of the population. The rapid response content takes a maximum of two minutes of interview time. The questions appear in the questionnaire for one or town collection periods (three to six months) and are asked of all CCHS respondents during that period, excluding the Territories.

4.5 Content included in data files

The survey produces different data files:

- One-year data file
- Two-year combined data
- rapid response data file (when applicable)

Table 4.2 provides clarification about the data files available for the 2017 and 2018 CCHS.

One-year data files

The survey produces data files every year. The 2017 annual file includes respondents from the 2017 data collection and variables from the core and theme content, as well as optional content. The territories are excluded from the one-year data file as the sample is only representative of the territories after two years.

Two-year data files

With each release of an even year data file, for example 2018, a file combining two years of data is released. The next two year file is scheduled to be released in 2019, and will include both the 2017 and 2018 reference years of collection.

The two-year data file includes all respondents and questions that were in the survey over the two year reference period. Unless otherwise specified, it comprises the questions from the annual core content and two-year theme content, as well as optional content that was selected over the two year period. The one-year theme and optional content selected for only one year are not available in the two-year data file.

Table 4.2 Content components for the 2017 and 2018 data files

[3]. Unless all provinces and territories in Canada select an optional module in the same collection period, which has never happened to date.

Files		Annual core content	2017 one year theme content	2018 one year theme content	2017-2018 two-year theme content	Optional content[4]
2017	**Main**	Yes	Yes[5]	N/A	Yes[5]	Yes
2018	**Main**	Yes	N/A	Yes[5]	Yes[5]	Yes
2017-2018	**Main**	Yes	No	No	Yes	Yes

4. Optional content will be included in the 2017-2018 data file if it is asked of respondents in a province or territory during the two year period. Otherwise, it will only be included in the file of the year in which it was collected. Note that if a one-year theme content module is selected for the optional content of a jurisdiction during the second year, the module will be included in the two-year data file and will be processed as optional content.
5. One-year theme content is not available for the territories unless they have selected that module in their optional content selection for the other year.

5. SAMPLE DESIGN

5.1 Target population

The CCHS covers the population 12 years of age and over living in the ten provinces. The three territories are covered over two years of collection. Excluded from the survey's coverage are: persons living on reserves and other Aboriginal settlements in the provinces; full-time members of the Canadian Forces; youth aged 12 to 17 living in foster homes; the institutionalized population; and persons living in the Quebec health regions of Nunavik and Terres-Cries-de-la-Baie-James. Altogether, these exclusions represent less than 3% of the target population.

5.2 Health regions

For administrative purposes, each province is divided into health regions (HR) and each territory is designated as a single HR. Statistics Canada is sometimes asked to make minor changes to the boundaries of some of the HRs to correspond to the geography of the Census, or to better account for the health data needs determined by the new geographic boundaries. For CCHS 2017, data was collected in over 100 HRs in the ten provinces, as well as in one HR per territory (Appendix C).

5.3 Sample size and allocation

To provide reliable estimates for each HR given the budget allocated to the CCHS component, it was determined that the survey should consist of a sample of nearly 130,000 respondents over a period of 2 years. Starting in 2015, the samples for the youth population aged 12 to 17 and the adult population aged 18 and over are treated separately. Based on the historical sample sizes, 120,000 of the 130,000 respondents were allocated to the adults' population and about 10,000 to the youth population.

Although producing reliable estimates for each HR was a primary objective, the quality of the estimates for certain key characteristics at the provincial level was also deemed important. Therefore, the sample allocation strategy, consisting of three steps, gave relatively equal importance to the HRs and the provinces. In the first step, a minimum size of 500 respondents per HR was imposed. This is considered the minimum for obtaining a reasonable level of data quality. However, due to response burden, a maximum sampling fraction of 1 out of 20 dwellings was imposed to avoid sampling too many dwellings in smaller regions also targeted by other surveys. Note that very few HRs have a size lower than 500 due to limit of the sampling fraction. In the first step, provinces and territories are treated separately. A sample of 117,000 respondents is allocated to the provinces and 3,000 respondents are allocated to the territories. Within each group, the sample is allocated using a 0.75 power allocation based on the population size. Table 5.1 gives the targeted sample sizes for 2017 and 2017-2018. Note that the sample size and allocation for the territories only allow for representativeness over a two year period.

Table 5.1 Number of health regions and targeted sample sizes by province/territory, 2017 and 2017-2018

		Adults		Youth	
Province	Number of HRs	Target sample size 2017	Target sample size 2017-2018	Target sample size 2017	Target sample size 2017-2018
Newfoundland and Labrador	4	1,586	3,172	157	313
Prince Edward Island	1	1,058	2,115	111	221
Nova Scotia	4[6]	2,486	4,971	193	385
New Brunswick	7	2,091	4,182	164	328
Quebec	16	12,402	24,804	948	1,896
Ontario	36	18,177	36,354	1,516	3,032
Manitoba	5	2,902	5,803	261	522
Saskatchewan	11	2,564	5,128	226	453
Alberta	5	6,992	13,984	587	1,175
British Columbia	16	8,194	16,389	629	1,259
Yukon	1	518	1,037	99	199
Northwest Territories	1	564	1,129	99	199
Nunavut	1	417	834	99	199
Canada	108	59,951	119,901	5,090	10,181

Then, within each province, the sample is allocated to the HRs using a 0.35 power allocation based on the size of the population of the HR. Note that the sample sizes obtained are then inflated to compensate for non-response and out-of-scope units. This inflated sample size is called the raw sample size and corresponds to the number of dwellings that need to be sampled to obtain the required number of respondents. Sample sizes by HR are available in Appendix D for 2017.

5.4 Frames, household sampling strategies

The CCHS sample is selected using different frames according to the age group. For the adult population (18 years and older), the sample of households is selected from an area frame. For the youth population (12 to 17 years old) a list frame is used to select persons.

5.4.1 Sampling of households from the area frame for the adult population

The area frame used by the Canadian Labour Force Survey (LFS) is used as a sampling frame for the adult population. The sampling plan of the LFS is a two-stage stratified cluster design in which the dwelling is the final sampling unit. In the first stage, homogeneous strata are formed and

[6] One change in Nova Scotia HR boundaries: 4 zones instead of 9.

independent samples of clusters are drawn from each stratum. In the second stage, dwelling lists are prepared for each cluster, and dwellings, or households are selected from these lists.

For the purpose of the LFS plan, geographic or socio-economic strata are created within each province. Within the strata, between 150 and 250 dwellings are grouped together to create clusters. Some urban centres have separate strata for apartments or for census Dissemination Areas (DA) to pinpoint households with high income, immigrants and aboriginal people.

Once the new clusters are listed, the sample is obtained using a systematic sampling of dwellings. The sample size for each systematic sample is called the "yield". As the sampling rates are determined in advance, there is frequently a difference between the expected sample size and the numbers that are obtained. The yield of the sample, for example, is sometimes excessive. This can particularly happen in sectors where there is an increase in the number of dwellings due to new construction. To reduce the cost of collection, an excessive output is corrected by eliminating, from the beginning, a part of the units selected and by modifying the weight of the sample design. This change is dealt with during weighting.

Due to the specificity of the CCHS, some modifications had to be incorporated in this sampling strategy. To obtain an annual sample of about 60,000 respondents for a given year of CCHS, about 100,000 dwellings had to be selected from the area frame to account for vacant dwellings and non-responding households. Each month, the LFS design provides approximately 60,000 dwellings distributed across the various economic regions in the ten provinces, whereas the CCHS required 100,000 dwellings distributed across the HRs, which have different geographic boundaries from those of the LFS economic regions. Overall, the CCHS required a lower number of dwellings than those generated by the LFS selection mechanism, which corresponds to an average *adjustment factor* of 0.6 (60,000/100,000). However, since the adjustment factors varied at the HR level, certain adjustments were required.

The changes made to the selection mechanism in the regions varied depending on the size of the adjustment factors. For HRs that had a factor smaller than or equal to 1, the number of Primary Sampling Units (PSUs) selected was reduced if necessary. For example, if the factor was 0.5, then only 3 PSUs were selected in each stratum instead of the usual number of 6 PSUs. For those HRs with a factor greater than 1 but smaller than or equal to 2, the sampling process of dwellings within a PSU was repeated for a subset of the selected PSUs that were part of the same HR. For example, if the factor was 1.6 then the selection of dwellings within a PSU was repeated for 4 of the 6 PSUs in all strata of that HR. When it was necessary to have a repeated selection of dwellings within a PSU and there were no more dwellings available in that PSU, then another PSU was selected. When the factor was greater than 2, the sampling process of dwellings was repeated among other PSUs that were part of the same HR[7].

7. To reduce listing costs, the sampling process of dwellings was repeated up to 3 times within PSUs already selected in urban areas only. These cases were exceptions, however.

Finally, when the number of dwellings available in the selected PSUs was greater than the requested number of dwellings for a given HR, a sub-sample of dwellings was selected. This process is called *stabilization*.

Sampling of households from the area frame in the three territories

For operational reasons, the LFS area frame sample design for the three territories was different. For each territory, the larger communities each have their own stratum while smaller communities are grouped into strata based on various characteristics (population, geographical information, proportion of Inuit and/or Aboriginal persons, and median household income). The LFS defined six design strata in the Yukon, ten in the Northwest Territories and ten in Nunavut. For strata consisting of a group of communities, the first stage of selection consisted of randomly selecting one community with a probability proportional to population size within each design stratum. Then, within the selected community, the second stage consisted of selecting households using the same sampling strategy as the one described above. The CCHS selected its sample from the same communities sampled by the LFS, while ensuring that different dwellings were selected. If too many or too few dwellings were available for a community within a stratum, another community was selected for the CCHS. For larger communities with their own stratum, only one stage design was necessary where households were selected directly using the same sampling strategy described above.

It is worth mentioning that the frame for the CCHS covered 94% of the targeted population in the Yukon[8], 96% in the Northwest Territories and 93% in Nunavut[9].

5.4.2 Sampling of persons from the list frame for the youth population

To sample persons for the youth population between the ages of 12 and 17 years old, the CCHS uses a list frame created from the Canadian Child Tax Benefits (CCTB) files. The CCTB files contain a list of all program beneficiaries with their names, addresses and phone numbers. This list is used to select directly the youths who will be interviewed over the phone.

5.5 Sample allocation over the collection periods

In order to balance interviewer workload and to minimize possible seasonal effects on estimates, the initial sample size for each frame is allocated equally over the four 3-month collection periods.

In the area frame (adults), each PSU selected within each HR was randomly assigned to a collection period accounting for a number of constraints related to field operations or weighting, while maintaining a uniform size for each period.

[8] In Yukon, after the 2016 Census of Population, the coverage was expanded to represent 94% of the targeted population. Before it was 92%.

9. In Nunavut, after the 2016 Census of Population, the coverage was expanded to represent 93% of the targeted population. Before it was 92%.

For the sample of youth selected from a list, independent samples were selected in each collection period. This strategy ensures that each sample is representative of the in scope Canadian population aged 12 to 17 years old for each three-month period.

5.6 Sampling of interviewees

Since the sample for the youth population is selected from a list of individuals, the selection of the interviewee is automatically handled at sampling and no extra step is required at collection. The selection of an interviewee at collection is now only required for the adult population.

The selection strategy that was adopted accounted for user needs, cost, design efficiency, response burden and operational constraints. For a household selected as part of the 18+ sample, one person is selected per household using varying probabilities taking into account the age and the household composition. The selection probabilities resulted from simulations using various parameters in order to determine the optimal approach without causing extreme sampling weights.

Table 5.3 gives the selection weight multiplicative factors used to determine the probabilities of selection of individuals in sampled households by age group. For example, for a three-person household formed of two people between the ages of 35 and 64 and one 19-year-old, the 19-year-old would have a 1/2 chance of being selected (i.e., 20/(20+10+10)) while each of the 35 to 64 year olds would have a 1/4 chance of being selected. To avoid extreme sampling weights, there is one exception to this rule: if the size of the household is greater than or equal to 5 or if the number of 18-19 year olds is greater than or equal to 3 then the selection weight multiplicative factor equals 1 for each individual in the household. Consequently, all people in that household have the same probability of being selected.

Table 5.3 Selection weight multiplicative factors for the person-level sampling strategy by age

Selection Weight Multiplicative Factors				
Age	**18-34**	**35-49**	**50-64**	**65+**
Factors	20	10	10	20

6. DATA COLLECTION

6.1 Computer-assisted interviewing

The CCHS uses two separate CAI applications to collect data, one for telephone interviews (CATI) and one for personal interviews (CAPI). This was done in order to customise each application's functionality to the type of interview being conducted. Each application consisted of an entry component, a health content, and an exit component.

Between January and December 2017, approximately 58,600 valid interviews were conducted using CAI. Approximately 26% of these completed cases were conducted in person using CAPI, and the other 74% were conducted over the phone using CATI.

CAI offers two main advantages over other collection methods. First, CAI offers a case management system and data transmission functionality. This case management system automatically records important management information for each attempt on a case and provides reports for the management of the collection process. CAI also provides an automated call scheduler, i.e. a central system to optimise the timing of call-backs and the scheduling of appointments used to support CATI collection.

The case management system routes the questionnaire applications and sample files from Statistics Canada's main office to regional collection offices (in the case of CATI) and from the regional offices to the interviewers laptops (for CAPI). Data returning to the main office takes the reverse route. To ensure confidentiality, the data is encrypted before transmission. The data are then decrypted when they are on a separate secure computer with no remote access.

Second, CAI allows for custom interviews for every respondent based on their individual characteristics and survey responses. This includes:

- questions that are not applicable to the respondent are skipped automatically
- edits to check for inconsistent answers or out-of-range responses are applied automatically and on-screen prompts are shown when an invalid entry is recorded. Immediate feedback is given to the respondent and the interviewer is able to correct any inconsistencies.
- question text, including reference periods and pronouns, is customised automatically based on factors such as the age and sex of the respondent, the date of the interview and answers to previous questions.

6.2 CCHS application development

Entry and exit components were developed for CATI and CAPI interviewing. These components contain standard sets of questions designed to guide the interviewer through contact initiation, collection of important sample information, respondent selection and determination of cases status. The health content consists of the health modules themselves and made up the bulk of the applications. This includes common modules asked of all respondents and optional modules which differed by provinces and territories. Each application underwent three stages of testing: block, integrated and end to end.

Block level testing consists of independently testing each content module or "block" to ensure skip patterns, logic flows and text, in both official languages, are specified correctly. Skip patterns or logic flows across modules are not tested at this stage as each module is treated as a standalone questionnaire. Once all blocks are verified by several testers, they are added together along with entry and exit components into integrated applications. These newly integrated applications are then ready for the next stage of testing.

Integrated testing occurs when all of the tested modules are added together, along with the entry and exit components, into an integrated application. This second stage of testing ensures that key demographic information such as age, sex, and economic family status are passed from the entry to the health content and exit components of the applications. It also ensures that variables affecting skip patterns and logic flows are correctly passed between modules within the health content. Since, at this stage the applications essentially function as they will in the field, all possible scenarios faced by interviewers are simulated to ensure proper functionality. These scenarios test various aspects of the entry and exit components including, establishing contact, collecting contact information, determining whether a case is in scope, rostering households, creating appointments and selecting respondents. The applications are also tested to ensure that during an interview, correct modules are triggered reflecting health region optional content selections.

End-to-end testing occurs when the fully integrated applications are placed in a simulated collection environment. The applications are loaded onto computers that are connected to a test server. Data is then collected, transmitted and extracted in real time, exactly as it would be done in the field. This last stage of testing allows for the testing of all technical aspects of data input, transmission and extraction for each of the CCHS applications. It also provided a final chance of finding errors within the entry, health content and exit components.

6.3 Interviewer training

Project managers, senior interviewers and interviewers from regional collection offices were sent self-study training packages before the start of collection. These packages were prepared by the CCHS project team and were used by existing experienced CCHS interviewers to reinforce their previous training. Project managers and senior interviewers also conducted customised training sessions for new CCHS interviewing staff as needed. There were also specific training sessions to deal with various topics related to CCHS collection on a monthly basis. The focus of the training sessions were to get interviewers comfortable using the CCHS 2017 applications, and familiarise interviewers with survey content and to introduce interviewers to interviewing procedures specific to the CCHS. The training focused on:

- goals and objectives of the survey including a focus on the survey redesign
- survey methodology
- application functionality
- review of the questionnaire content and exercises with an emphasis on significant content changes
- interviewer techniques for maintaining response - complete exercises to minimise non-response

- use of mock interviews to simulate difficult situations and practise potential non-response situations
- survey management
- transmission procedures

One of the key aspects of the training was a focus on minimizing non-response. Exercises to minimise non-response were prepared for interviewers. The purpose of these exercises was to have the interviewers practice convincing reluctant respondents to participate in the survey. There was also a series of refusal avoidance workshops given to the senior interviewers responsible for refusal conversion in each regional collection office.

6.4 The interview

Sample units were selected from an area frame for the population aged 18 and older, and the Canadian Child Tax Benefit (CCTB) frame for the population aged 12 to 17. Approximately 75% of the respondents selected from the area frame and all of the respondents selected from the CCTB frames were interviewed from centralised call centres using CATI. The CATI interviewers were supervised by a senior interviewer located in the same call centre. About 25% of the area frame respondents were interviewed by decentralised field interviewers using CAPI. While in some situations field interviewers were permitted to complete some or part of an interview by telephone, roughly three quarters of these interviews were conducted exclusively in person. CAPI interviewers worked independently from their homes using laptop computers and were supervised from a distance by senior interviewers. The variable CASETYPE on the microdata file indicates which frame a case was selected from, as well as the collection mode it was completed in. The following shows the breakdown of CASETYPE:

- CASETYPE = 0: case was selected from area frame and completed in CAPI
- CASETYPE = 1: case was selected from CCTB frame (12-17 years old) and completed in CATI
- CASETYPE = 2: case was selected from area frame and completed in CATI

CAPI interviewers were trained to make an initial personal contact with each sampled dwelling. In cases where this initial visit resulted in non-response, telephone follow-ups were permitted. The variable ADM_N040 on the microdata files indicates whether the CAPI interview (CASETYPE=0) was completed face-to-face, by telephone or using a combination of the two techniques.

In all selected dwellings, a knowledgeable household member was asked to supply basic demographic information on all residents of the dwelling. One member of the household was then selected for a more in-depth interview, which is referred to as the health content interview.

To ensure the quality of the data collected, interviewers were instructed to make every effort to conduct the interview with the selected respondent in privacy. In situations where this was unavoidable, the respondent was interviewed with another person present. Flags on the microdata

files indicate whether somebody other than the respondent was present during the interview (ADM_N045) and whether the interviewer felt that the respondent's answers were influenced by the presence of the other person (ADM_N050).

To ensure the best possible response rate attainable, many practices were used to minimise non-response, including:

a) Introductory letters

Before the start of each collection period, introductory letters and brochures explaining the purpose of the survey were sent to the sampled households. These explained the importance of the survey and provided examples of how CCHS data would be used.

b) Initiating contact

Interviewers were instructed to make all reasonable attempts to obtain interviews. When the timing of the interviewer's call (or visit) was inconvenient, an appointment was made to call back at a more convenient time. If requests for appointments were unsuccessful over the telephone, interviewers were instructed to follow-up with a personal visit. If no one was home on first visit, a brochure with information about the survey and intention to make contact was left at the door. Numerous call-backs were made at different times on different days.

c) Refusal conversion

For individuals who at first refused to participate in the survey, a letter was sent from the nearest Statistics Canada Regional Office to the respondent, stressing the importance of the survey and the household's collaboration. This was followed by a second call (or visit) from a senior interviewer, a project supervisor or another interviewer to try to convince respondent of the importance of participating in the survey.

d) Language barriers

To remove language as a barrier to conducting interviews, each of the Statistics Canada Regional Offices recruited interviewers with a wide range of language competencies. When necessary, cases were transferred to an interviewer with the language competency needed to complete an interview.

e) Youth interviews

In 2017, interviewers needed to obtain verbal permission from parents/guardians to interview youths between the ages of 12 to 15 who were selected for interviews. This information was collected in the Parental/Guardian Consent (PGC) block in the survey entry component. Several procedures were followed by interviewers to alleviate potential parental concerns and to ensure a completed interview. Interviewers carried with them a card entitled "Note to parents / guardians about interviewing youths for the Canadian Community Health Survey". This card explained the purpose of collecting information from youth, lists the subjects to be covered in the survey, asks for permission to share and link the obtained information and explains the need to respect a youth's right to privacy and confidentiality.

If a parent/guardian asked to see the actual questions; interviewers were instructed to either show the survey questions, or if the interviewer was being conducted by phone, to immediately have the regional office send a copy of the questionnaire.

If privacy could not be obtained to interview the selected youth either in person or over the phone (another person listening in) the interview was coded a refusal. However, for CAPI interviews, if privacy could not be obtained to interview the selected youth, the interviewer was able to propose to the parent/guardian that the interviewer read the questions out loud and the youth enter their answers directly on the computer.

The Person Most Knowledgeable (PMK) block collected household level information found at the end of the survey (Insurance coverage, Food Security, Income and Administration) from the most knowledgeable person in the household. In 2017, this block is initiated when the selected respondent is between the ages of 12 to 17. The block formalizes the process of identifying a person in the household who is likely better able to answer these household level questions than the young selected respondent. If a PMK is found, then the interview moves from the younger selected respondent between the ages of 12 and 17, to a household member who finishes the rest of the interview after the PMK block.

f) Proxy interviews

In cases where the selected respondent was, for reasons of physical or mental health, incapable of completing an interview, another knowledgeable member of the household supplied information about the selected respondent. This is known as a proxy interview. While proxy interviewees were able to provide accurate answers to most of the survey questions, the more sensitive or personal questions were beyond the scope of knowledge of a proxy respondent. This resulted in some questions from the proxy interview being unanswered. Every effort was taken to keep proxy interviews to a minimum.

Since 2010, the Proxy interview (GR) block has been modified to prompt the interviewer to specifically identify whether the proxy interview is being conducted due to a physical or mental condition. Interviewers are then asked to record the specific condition for either case. The variable ADM_PRX indicates whether a case was completed by proxy.

6.5 Field operations

The 2017 sample was divided on a yearly basis into 4 non-overlapping three-month collection periods. Regional collection offices were instructed to use the first 4 weeks of each collection period to resolve the majority of the sample, with next 8 weeks being used finalise the remaining sample and to follow up on outstanding non-response cases. All CATI cases were to have been attempted by the second week of each collection period. Sample files were sent approximately two weeks before the start of each collection period to centralised collection offices. A series of dummy cases were included with each CAPI sample. These cases were completed by senior interviewers for the purposes of ensuring that all data transmission procedures were working through the collection cycle. Once the samples were received, project supervisors were responsible for

planning CAPI interviewer assignments. Wherever possible, assignments were generally no larger than 15 cases per interviewer.

Transmission of cases from each of the CATI offices to head office was the responsibility of the regional office project supervisor, senior interviewer and the technical support team. These transmissions were performed nightly and sent all completed cases to Statistics Canada's head office. Completed CAPI interviews were transmitted daily from the interviewer's home directly to Statistics Canada's head office using a secure telephone transmission.

For final response rates, refer to Appendix E.

6.6 Quality control and collection management

During the collection year, several methods are used to ensure data quality and to optimize collection. These included using internal measures to verify interviewer performance and the use of a series of ongoing reports to monitor various collection targets and data quality.

A system of validation was used for CAPI cases whereby interviewers had their work validated on a regular basis by the Regional Office. Each collection period, randomly selected cases were flagged in the sample. Regional office managers and supervisors created lists of cases to be validated. These cases were handed to the validation team who then contacted households to verify that a legitimate interview took place. Validation procedures generally occurred during the first few weeks of a collection period to ensure that any issues were detected promptly. Interviewers were provided feedback by their supervisors on a regular basis.

Since 2011, an additional quality control system was introduced for CAPI interviews. Upon obtaining consent from the respondents, specific sections of the CAPI interviews were recorded. These recordings were transmitted back to the regional offices, and then randomly chosen for analysis. CATI interviewers were also randomly chosen for validation. Validation in the CATI collection offices consisted of senior interviewers monitoring interviews to ensure proper techniques and procedures (reading the questions as worded in the applications, not prompting respondents for answers, etc.) were followed by the interviewer.

A series of reports were produced to effectively track and manage collection targets and to assist in identifying other collection issues.

Cumulative reports were generated at the end of each collection period, showing response, link, share and proxy rates for both the CATI and CAPI samples by individual health region. The reports were useful in identifying health regions that were below collection target levels, allowing the regional offices to focus efforts in these regions.

Using information obtained from the CAI applications, further analysis was done in head office in order to identify interviews that were completed below acceptable time frames. These short interviews were flagged, removed from the microdata and treated as non-response.

In 2014, a new collection management approach was implemented to improve the efficiency of the CATI data collection: Responsive Collection Design. Responsive Collection Design (RCD) is

a new collection strategy that allows the Regional Offices (ROs) to adjust data collection approaches throughout the collection period. In basic terms, it allows the ROs to group cases according to a number of factors, and then target the groups that we are interested in. In an ordinary CATI survey, the interviewer groups remain constant throughout collection. However, during Responsive Collection Design, these groups change during each phase of collection in order to maximize response rates, productivity and the representativeness of the collected sample.

With the 2015 CCHS redesign, a dynamic telephone number strategy was developed. Starting in 2015, the 18+ years old sample can have up to 2 phone numbers, while the CCTB samples can have up to 4 telephone numbers. This increases the number of available telephone numbers CATI interviewers can call to maximize their potential ability to reach the selected households or respondents. In addition, the functionality of transferring CATI (telephone) area frame cases to CAPI field interviewers was added. For cases that were contacted in CATI and reached the wrong address, had the wrong telephone number, or could not be reached, these cases would be transferred to the field CAPI interviewers who would then physically visit the listed address to try and collect a response. This ensures that all cases will have sufficient contact attempts.

7. DATA PROCESSING

7.1 Editing

Most editing of the data was performed at the time of the interview by the computer-assisted interviewing (CAI) application. It was not possible for interviewers to enter out-of-range values and flow errors were controlled through programmed skip patterns. For example, CAI ensured that questions that did not apply to the respondent were not asked.

In response to some types of inconsistent or unusual reporting, warning messages were invoked but no corrective action was taken at the time of the interview. Where appropriate, edits were instead developed to be performed after data collection at Head Office. Inconsistencies were usually corrected by setting one or both of the variables in question to "not stated".

7.2 Flows

When data collected in the field is returned to Head Office, any questions that were skipped will appear as blank on the preliminary data files. During processing, every blank variable is given one of two processing code values:

‘Not applicable’ - NA – For variables of length 1, the value for NA will simply be 6. Otherwise, these values will appear on the file as a string of 9’s, followed by a 6, according to the length of the numerical variable or text field.

The code NA is applied whenever a question is skipped because it does not apply to the respondent (a population exclusion). The two main reasons for this code being used are optional module variables for respondents in provinces where that content was not selected; and questions that are skipped by a flow in the questionnaire since it did not apply, as with the Maternal Experiences questions for male respondents.

‘Not stated’ – NS – For a variable of length 1, the value for NS will simply be 9. Otherwise, these will appear on the file as a string of 9’s, according to the length of the numerical variable or text field.

The code NS is applied whenever a question is skipped, but could have been asked. These are respondents that are in the universe, but without an actual response. The two main reasons for this code being used are questions that are skipped because the interview is being completed by proxy; and questions that are skipped because a preceding question was answered with a ‘Don’t Know’, ‘Refuse’, or else was itself ‘Not Stated’. For example, if a respondent refuses to answer CCC_Q095 (Do you have diabetes?), the questions that ask about the age of the diagnosis, insulin use, etc., are all set to ‘Not Stated’ since it is not known whether or not the respondent has diabetes.

7.3 Coding

Pre-coded answer categories were supplied for all suitable variables. Interviewers were trained to assign the respondent’s answers to the appropriate category.

In the event that a respondent's answer could not be easily assigned to an existing category, several questions also allowed the interviewer to enter a long-answer text in the "Other-specify" category. All such questions were closely examined in head office processing. For some of these questions, write-in responses were coded into one of the existing listed categories if the write-in information duplicated a listed category. In 2017, variables that were OSLA coded include INJ_050, INJ_065, INJ_075, INJ_080, MEX_105, MEX_120, MEX_195, STS_015 and PG_01. Other variables with other-specify responses for this cycle were not selected for coding. For all questions, the 'Other-specify' responses are taken into account when refining the answer categories for future cycles.

When write-in responses ("Other-specify") are coded into an existing category during head office processing, it is possible that other questions in the questionnaire would become in scope to the respondent. Since those questions were never asked during the interview, the missing answers were set to 'Not Stated' in processing. For example, in the Injuries (INJ) module, a question asks what the respondent was doing at the time of their injury (INJ_Q065). If the interviewer collected an answer in the "Other Specify" that indicated the respondent was working, then the variable INJ_065 would be coded to the category 'Working at a job or business (excluding travel to and from work)' and assigned a value of '04'. Had the interviewer used this category in the interview, the respondent would then get the Workplace injury (INW) block of questions if it were survey content selected as optional content. All of the questions in INW in this case will have been set to 'Not Stated'.

7.4 Creation of derived variables

To facilitate data analysis and to minimize the risk of error, a number of variables on the file have been derived using items found on the CCHS questionnaire. Derived variables generally have a "D", "G" or "F" in the fourth character of the variable name. In some cases, the derived variables are straightforward, involving the collapsing of response categories. In other cases, several variables have been combined to create a new variable. The *Derived Variables Documentation (DV)* provides details on how these more complex variables were derived. For more information on the naming convention, please go to Section 12.6.

7.5 Weighting

The principle behind estimation in a probability sample such as CCHS is that each person in the sample "represents", besides himself or herself, several other persons not in the sample. For example, in a simple random 2% sample of the population, each person in the sample represents 50 persons in the population. In the terminology used here, it can be said that each person has a weight of 50.

The weighting phase is a step that calculates, for each person, his or her associated sampling weight. This weight appears on the microdata files, and must be used to derive meaningful estimates from the survey. For example, if the number of individuals who smoke daily is to be estimated, it is done by selecting the records referring to those individuals in the sample having that characteristic and summing the weights entered on those records.

7.6 Income variables for 2017 – Linkage and Imputation

As in 2016, income variables in the 2017 CCHS master file are populated from one of three sources: tax records, respondent provided data, imputed data. Where respondents did not object to a link to their tax data and where a link to the appropriate tax records could be found, the income source variables (INC_005A-O, INC_Q025A-O), the main source of income (INC_015, INC_035), if they received a supplement for people with a disability (INC_010, INC_030) and the total income amounts (INC_021, INC_041) were determined from tax records. Where linkage to tax data was not feasible, respondent data was used. If neither linked tax data nor respondent reported data was available, an imputation of the total personal and total household income variables was conducted. The sources of incomes (INC_005A-O, INC_010, INC_015, INC_Q025A-O, INC_030, INC_035) were not imputed.

The variable INCFIMPP indicates the source used for the personal income variables and INCFIMPH indicates the source for household income variables. For more information concerning incomes on the master file, please refer to the document "Income variables and sources in the Canadian Community Health Survey master file" available under the 'Documentation' section of the 2017 CCHS Microdata File User Guide.

Definitions, data sources and methods page on the Statistics Canada website or contact Client Services (613-951-1746; fax: 613-951-0792; statcan.HD-DS.statcan@canada.ca). For share file users, please read, "Income variables and sources in the Canadian Community Health Survey share file".

7.7 Postal code imputation

A respondent's postal code is employed, using the Postal Code Conversion File, to derive the rest of the geographical variables that are available on the CCHS data file. It is therefore important that all respondents have a valid postal code. If a respondent's postal code is missing or invalid, it is usually imputed through a donor imputation process, although other imputation methods are sometimes used. The donor is chosen from the same geographical area, with as much precision as possible, as the unit with the missing or invalid postal code.

For the 2017 CCHS, approximately 2% of respondents from the area frame (those aged 18 and over) had their postal codes imputed. For respondents from the CCTB frame (aged 12-17), approximately 0.5% had their postal codes imputed in 2017.

8. WEIGHTING

In order for estimates produced from survey data to be representative of the covered population, and not just the sample itself, users must incorporate the survey weights in their calculations. A survey weight is given to each respondent included in the final sample. This weight corresponds to the number of persons in the entire population that are represented by the respondent.

As described in Section 5, the CCHS uses two sampling frames for its sample selection: an area frame for the Canadian population aged 18 and over, and a frame of telephone numbers from Canada Child Tax Benefit (CCTB) records for the 12-17 population.

The weighting strategy treats both the area and CCTB frames independently to come up with separate person-level weights for each of the frames used. The person-level weights from the two frames are then combined into a single set of weights, jointly undergo a couple more adjustments (including being matched to known population totals), and become the final person-level weights.

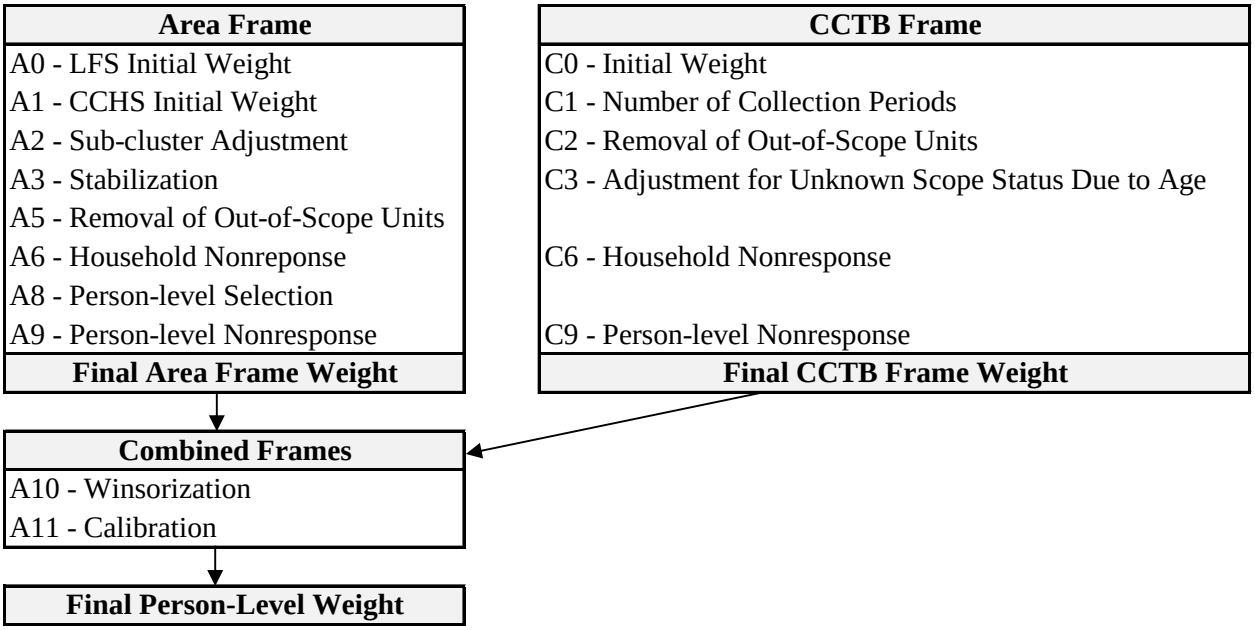
8.1 Overview

Units from both the area and CCTB frames are treated separately up to and including the adjustment for person-level nonresponse. The following sections describe the weighting process for the provinces. Sub-section 8.2 provides details on the weighting strategy for the area frame, while sub-section 8.3 deals with the strategy for the CCTB frame. The final weighting steps of Winsorization and calibration, which involve both frames, are discussed in sub-section 8.4.

Following the redesign of the CCHS in 2015, due to operational constraints, collection of a representative sample will be spread over a two year period. This means that estimates in the three territories will only be representative over two years. As such, respondents in the territories are omitted from the one-year files and will only be included on the two-year files (discussed in sub-section 8.6). There are some particularities associated with the weighting process for the territories, and this will be discussed in more detail in documentation of two-year CCHS files.

Diagram A presents an overview of the most important adjustments that are part of the weighting strategy. Adjustments applied to units from the area frame are enumerated from *A0* to *A9*, while adjustments applied to units from the CCTB frame are enumerated from *C0 to C9*. The adjustments once the frames have been combined are called *A10* and *A11*.

Diagram A Weighting Strategy Overview



8.2 Weighting of the area frame sample

A0 – LFS Initial Weight

The weighting process on the area frame sample begins with a weight provided by the Labour Force Survey (LFS). This weight is based on the LFS design since the CCHS area frame sample design is based on the LFS. The LFS design consists of a sample of dwellings within clusters selected from LFS strata, and the LFS initial weights reflect the probabilities of selection within LFS strata.

A1 – CCHS Initial Weight

In the initial adjustment A1, the LFS initial weight is adjusted to take into consideration the fact that the CCHS selects a sample of sufficient size for estimation at the Health Region level. To do so, the CCHS selects a different number of clusters than the LFS and can repeat the sampling of dwellings within the selected clusters. The resulting weight is called weight A1. For more details about the selection mechanism, as well as a more complete definition of LFS strata and clusters, refer to Statistics Canada (2017)[10].

[10] Statistics Canada, 201y. *Guide to the Labour Force Survey*, Statistics Canada. Cat. No. 71-543-G.

A2 – Sub-cluster Adjustment

In clusters that experience significant growth, a sub-sampling methodology is used to ensure that the workload of the interviewers is kept at a reasonable level. This can consist of sub-sampling from the selected dwellings, dividing the cluster into sub-clusters, or reclassifying the cluster as a stratum and creating new clusters within the stratum. In all these cases, a sub-sample adjustment is calculated and applied to the CCHS weight. This adjustment is applied to weight A1 to produce weight A2. Again, more information can be found in the LFS documentation (Statistics Canada (2017).

A3 – Stabilization

In some HRs, the increase in the sample size, as described in section 5, results in a larger sample than necessary. Stabilization is used to bring the sample size back down to the desired level. In order to minimize the dispersion of weights and thus reduce variability of estimates, stabilization is done in one of two ways within each HR. If the initial weights in each cluster within an HR are the same, the process consists of randomly sub-sampling dwellings at the HR level from the dwellings originally selected within each cluster. However, if the initial weights are not the same in each cluster within an HR, the process is done iteratively. First, the cluster with the lowest initial weight is selected and one unit from that cluster is removed. Then the weights are recalculated to adjust for the removed unit. Again the cluster with the lowest weight is selected and one unit is removed. This process is repeated until enough units have been removed. An adjustment factor representing the effect of this stabilization is calculated in order to adjust the probability of selection appropriately. This factor, multiplied by weight A2, produces weight A3.

A5 – Removal of Out-of-Scope Units

Among all dwellings sampled, a certain proportion are identified during collection as being out-of-scope. Dwellings that are demolished or under construction, vacant, seasonal or secondary, as well as institutions are examples of out-of-scope cases for the CCHS. Dwellings in which all residents are out-of-scope, for example, all are full-time members of the Armed Forces or visitors to Canada are also out-of-scope. All these out-of-scope dwellings and their associated weights are simply removed from the sample, as it is assumed that the weighted out-of-scope dwellings in the sample are representative of out-of-scope dwellings in the population. For dwellings whose scope status is unknown (not enough information was gathered to determine scope status), their weights are reduced by the out-of-scope rate of their HR to account for some proportion of them being out-of-scope. This leaves a sample that is representative of in-scope dwellings or households. In-scope dwellings maintain the same weight as in the previous step, which is now called weight A5.

A6 – Household Nonresponse

During collection, a certain proportion of sampled households inevitably result in nonresponse. This usually occurs when a household refuses to participate in the survey, provides unusable

data, or cannot be reached for an interview. Weights of the nonresponding households are redistributed to responding households within response homogeneity groups (RHGs), which are formed within province. In order to create the response homogeneity groups, a scoring method based on logistic regression models is used to determine the propensity to respond and these response probabilities are used to divide the sample into groups with similar response properties. The information available for nonrespondents is limited so the regression models use characteristics such as the collection period, geographic information and variables from Statistics Canada's Household Survey Frame, as well as paradata or process data, which includes the number of contact attempts, the time/day of attempt, and whether the household was called on a weekend or weekday. Separate models are created for dwellings that were contacted by telephone and for dwellings that were contacted in-person. An adjustment factor is calculated within each response homogeneity group as follows:

$$\frac{Sum\ of\ weights\ A5\ for\ all\ households}{Sum\ of\ weights\ A5\ for\ all\ responding\ households}$$

Weight A5 is multiplied by this adjustment factor to produce weight A6 for the responding households. Non-responding households are dropped from the process at this point.

A8 – Person-level Selection

Since persons are the desired sampling units, the household-level weights computed to this point need to be converted to the person level. This weight is obtained by multiplying the household weight by the inverse of the probability of selection of the person selected in the household. This gives the weight A8. As mentioned earlier, the probability of selection for an individual changes depending on the number of people in the household and the ages of those individuals (see Section 5.6 for more details).

A9 – Person-level Nonresponse

A CCHS interview for area frame dwellings can be seen as a two-part process. First, the interviewer gets the complete roster of the people within the household. Second, the selected person is interviewed. In some cases, interviewers can only get through the first part, either because they cannot get in touch with the selected person, or because that selected person refuses to be interviewed. Such individuals are defined as person nonrespondents and an adjustment factor must be applied to the weights of person respondents to account for this nonresponse. Using the same methodology that is used in the treatment of household nonresponse, the adjustment is applied within response homogeneity groups. In this process, the scoring method is used to define a response probability based on characteristics available for both respondents and non-respondents. All characteristics collected when creating the roster of household members are available for the estimation of the response probabilities as well as geographic

information and some paradata. The probabilities are grouped into response homogeneity groups and the following adjustment factor is calculated within each group:

$$\frac{Sum\ of\ weights\ A8\ for\ all\ selected\ persons}{Sum\ of\ weights\ A8\ for\ all\ responding\ selected\ persons}$$

Weight A8 for responding persons is multiplied by the above adjustment factor to produce weight A9. Nonresponding persons are dropped from the weighting process from this point onward.

8.3 Weighting of the CCTB frame sample

C0 –Initial Weight

The initial design weight is defined as the inverse of the probability of selection and is computed separately for the CCTB and area frame samples since the method of selection differs between these two frames. For the CCTB frame, the selection of children is done from the CCTB list frame within each CCHS stratum.

For the CCTB frame, children are randomly selected among those assigned to the specific HR. The probability of selection corresponds to the ratio of the number of sampled units to the number of children on the list within the HR. The ratio is based on the frame available and the number of units selected for the particular three-month collection period. The probability of selection can therefore change depending on sample allocation and frame updates. The inverse of these probabilities represents the initial weight C0.

C1 – Number of Collection Periods

Samples are drawn every three months on the CCTB frame. Each of these samples comes with an initial weight that allows each sample to be representative of the population at the HR level. To ensure that the total sample represents the population only once, an adjustment factor is applied to reduce the weights of each three-month sample. The adjustment factor applied to each three-month sample is equal to the inverse of the number of samples being combined (i.e. the number of collection periods). Following this adjustment, the entire list frame sample corresponds to the average population over the entire combined collection period. The initial weights (C0) are multiplied by this adjustment factor to produce weight C1.

C2 - Removal of Out-of-Scope Units

Selected children who are full-time members of the Canadian Armed Forces, are part of a First Nations reserve, or are deceased, are all examples of out-of-scope cases for the CCTB frame. Similar to the methods used on the area frame, these cases are simply removed from the process, leaving only in-scope children in the sample. These in-scope children keep the same weight as in the previous step, now called weight C2.

C3 – Adjustment for Unknown Scope Status Due to Age

Some of the selected children have birthdays that fall within a collection period. For a proportion of these children, their birthday can change their scope status due to age. Some selected children may be eleven at the beginning of the collection period and therefore out-of-scope, but become in-scope when they turn twelve. Other children may be in-scope at seventeen but become out-of-scope when they turn eighteen. The age of these children, and therefore scope status, can be resolved (and treated accordingly) when contact is made and the age at time of contact is established. However, the scope status is unknown when children cannot be contacted or asked their age. They have their weights adjusted by their modeled probability of being in-scope had they been contacted and asked their age at contact. The rate of in-scope children turning 12 and 18 amongst known scope cases is used as the adjustment factor for the weights of these unknown scope cases. Those found out of scope due to their age have their weights set to zero similar to other out-of-scope units, in-scope children keep their weights and those with unknown scope and birthdays that could affect their scope have their weights adjusted, the resulting weights are called C3.

C6 – Household Nonresponse

The adjustment applied here to compensate for the effect of household nonresponse is identical to the one applied for the area frame units that were contacted by telephone (adjustment A6). For the CCTB frame the selection was done at the person-level, however a household response is defined as a household where the roster was completed, similar to the area frame. The adjustment factor calculated within each response homogeneity group is obtained as follows:

$$\frac{\textit{Sum of weights } C3 \textit{ for all households}}{\textit{Sum of weights } C3 \textit{ for all responding households}}$$

The weight C3 of responding households is multiplied by this adjustment factor to produce the weight C6. Nonresponding households are removed from the process at this point.

C9 – Person-level Nonresponse

The adjustment applied here to compensate for the effect of person nonresponse is identical to the one applied for the area frame (adjustment A9). The adjustment factor calculated within each response homogeneity group is obtained as follows:

$$\frac{\textit{Sum of weights } C6 \textit{ for all persons}}{\textit{Sum of weights } C6 \textit{ for all responding persons}}$$

The weight C6 of responding persons is multiplied by this adjustment factor to produce the weight C9. Nonresponding persons are removed from the process at this point.

8.4 Final weight adjustments with area and CCTB frames combined

At this point in the process the sample from the two frames can be combined in anticipation of the final weight adjustments of Winsorization and calibration. It is not necessary to perform an additional weight adjustment in combining the frames because they cover mutually exclusive populations. The selection probability and subsequent weighting of a unit on a given frame is in no way impacted by the structure or selection of units from the other frame.

A10 – Winsorization

Following the series of adjustments applied to the respondents, some units may come out with extreme weights compared to other units of the same domain of interest. These units could have a large impact on the variance. In order to prevent this, the weight of these outlier units is adjusted downward using a “winsorization” trimming approach.

A11 – Calibration

The last step necessary to obtain the final CCHS weight is calibration (A11). Calibration is done using Statistics Canada’s generalized estimation system, G-EST, to ensure that the sum of the final weights corresponds to the population estimates defined at the HR level, for all 10 age-sex groups of interest. The five age groups are 12-17, 18-34, 35-49, 50-64, and 65+, for both males and females. Starting in 2009, additional controls at more detailed geographic levels were introduced for HRs where additional information is available. A minimum size of 20 respondents is required to calibrate at the HR by age and sex. When getting less than 20 respondents, some collapsing is done within province and / or within gender. At the same time, weights are adjusted to ensure that each collection period (three-month period) is equally represented within the sample. Note that the calibration is done using the most up to date geography and may not match the geography used at the time of sampling.

The population estimates are based on the 2011 Census counts and counts of birth, death, immigration and emigration since that time. The average of these monthly estimates for each of the HR-age-sex post-strata by collection period is used to calibrate. The weight A10 is adjusted using G-EST to obtain the final weight A11. Weight A11 corresponds to the ***final CCHS person-level weight*** and can be found on the master file with the variable name WTS_M.

8.5 Creation of a share weight

Along with the master file and PUMF which contain all CCHS respondents, a share file is created which contains only a portion (usually > 90%) of the original CCHS respondents. The individuals on this share file have agreed to share their data with certain partners. To compensate for the loss of some respondents from the file, the weights of these "sharers" must be adjusted by the factor:

$$\frac{Sum\ of\ weights\ A9\ (or\ C9)\ for\ all\ respondents}{Sum\ of\ weights\ A9\ (or\ C9) for\ all\ respondents\ agreeing\ to\ share\ their\ data}$$

Similar to the nonresponse adjustments, this factor is calculated within response homogeneity groups, where in this case, individuals with similar estimated propensity to share will be grouped together. This share adjustment is calculated separately for respondents from the area frame and from the CCTB frame. Weight A9 (or C9 for the CCTB respondents) for sharers is multiplied by the above adjustment factor to produce a share weight. As with the Master weighting process, the area and CCTB frames are combined at this point, and Winsorization and calibration (similar to adjustments A10 and A11) are applied to the share units. The final weight after these adjustments is called WTS_S.

8.6 Particular aspects of the weighting in the three territories

Starting in 2015-2016, the territories have been sampled on a two-year basis. Prior to 2015, each stratum was sampled in a year but to reduce collection costs, strata such as the capitals were sampled annually while other strata sampled just once every two years. Since characteristics of the population are quite variable from one community to the next, the sample is now only representative on a two year basis.

In addition to the two-year sampling scheme, the sampling frame used in the three territories is somewhat different from the one used in the provinces. Therefore, the weighting strategy is adapted to comply with these differences. This subsection summarises the changes applied to the steps described in subsections 8.1 to 8.5. For the area frame, an additional stage of selection is added in the territories where each territory is stratified into groupings of communities and one community is selected within each group. The capital of each territory forms a stratum on its own and is selected automatically at the first stage. This has an effect in the computation of the probability of selection, and therefore in the value of the initial weight (A0). The initial weight is calculated annually. Once the initial weight is calculated, the same series of adjustments (A1 to A5) is applied to the area frame units. The stabilisation adjustment (A3) in general is performed at the community level. The out-of-scope adjustment (A5) is performed at the community level and by mode of collection. Furthermore, since some strata are sampled annually and others biannually, an adjustment to combine the strata into one biannual sample is performed after the adjustment for out-of-scope. Household-level and person level nonresponse adjustment classes are built in the same way as for the provinces, using the same set of variables plus the variable year (year of collection).

Starting with the 2008 and 2007-2008 reference year products, controls have been put in place to ensure that the proportion of aboriginals and the proportion of individuals in the capital regions are controlled in the Northwest Territories and the Yukon. A similar control based on Inuit status was introduced for Nunavut. Starting in 2009, the proportion of individuals in the capital region

is controlled in Nunavut. These controls ensure that the proportion of the estimates represented by these different groups is consistent with proportions indicated by the 2011Census.

Prior to 2013, CCHS only covered the 10 largest communities in Nunavut. The population counts used in calibration were adjusted to take this undercoverage into consideration. Starting in 2013, CCHS increased its coverage to match the Labour Force Survey where 93% of the population is covered. Therefore, the population counts used for calibration are based on the total population and no longer adjusted for this undercoverage.

8.7 Weighting for a two-year file

The territories are weighted as specified in section 8.6. They are weighted to be representative of the two years. The provinces have been weighted at an annual level. When two years of data are combined to create a two-year file, new weights are calculated by halving the annual weights. This ensures that the sum of the final weights is equal to the average population size over the two years. In some cases the adjustment is a little more complex where groups had been collapsed in calibration for the one year files. For more information on combining multiple years, please refer to the article "Combining cycles of the Canadian Community Health Survey" published in the Statistics Canada Health Reports publication (82-003) at the following link:

https://www150.statcan.gc.ca/n1/pub/82-003-x/2009001/article/10795-eng.pdf

Note that combining cycles of CCHS data from before and after the redesign (e.g. combining 2014 and 2015 Annual files) is NOT recommended because of the numerous changes in content and methodology.

9. DATA QUALITY

9.1 Response rates for 2017

In total, 92,542 of the selected units in the 2017 CCHS were in-scope for the survey[11], out of which a response was obtained for 58,133 individuals, resulting in a response rate of 62.8%. Appendix E provides the counts of in-scope units, the counts of respondents and the resulting response rates, by province and health region. These figures are provided for the adults and for the youths separately, as well as for the complete CCHS sample.

11. Among the units selected, some are not in-scope for the survey. They are, for example, dwellings selected from the area frame but that turned out to be vacant, demolished or non-residential dwellings, or youths that have moved outside Canada. These units are identified during the data collection, otherwise they would have been excluded before the sample selection. These units are not considered in the calculation of response rates. This response rate includes the three territories. Data for the three territories will be available in the 2017-2018 CCHS.

9.2 Survey Errors

The estimates derived from this survey are based on a sample of individuals. Somewhat different figures might have been obtained if a complete census had been taken using the same questionnaire, interviewers, supervisors, processing methods, etc. than those actually used. The difference between the estimates obtained from the sample and the results from a complete count under similar conditions is called the *sampling error* of the estimate.

Errors which are not related to sampling may occur at almost every phase of a survey operation. Interviewers may misunderstand instructions, respondents may make errors in answering questions, the answers may be incorrectly entered on the computer and errors may be introduced in the processing and tabulation of the data. These are all examples of *non-sampling errors*.

9.2.1 Non-Sampling Errors

Over a large number of observations, randomly occurring errors will have little effect on estimates derived from the survey. However, errors occurring systematically will contribute to biases in the survey estimates. Considerable time and effort was made to reduce non-sampling errors in the CCHS. Quality assurance measures were implemented at each step of data collection and processing to monitor the quality of the data. These measures included the use of highly skilled interviewers, extensive training with respect to the survey procedures and questionnaire, and the observation of interviewers to detect problems. Testing of the CAI application and field tests were also essential procedures to ensure that data collection errors were minimized.

A major source of non-sampling errors in surveys is the effect of *non-response* on the survey results. The extent of non-response varies from partial non-response (failure to answer just one or some questions) to total non-response. Partial non-response to the CCHS was minimal; once the questionnaire was started, it tended to be completed with very little non-response. Total non-response occurred either because a person refused to participate in the survey or because the interviewer was unable to contact the selected person. Total non-response was handled by adjusting the weight of persons who responded to the survey to compensate for those who did not respond. See section 8 for details on the weight adjustment for non-response.

9.2.2 Sampling Errors

Since it is an unavoidable fact that estimates from a sample survey are subject to sampling error, sound statistical practice calls for researchers to provide users with some indication of the magnitude of this sampling error. The basis for measuring the potential size of sampling errors is the standard deviation of the estimates derived from survey results. However, because of the large variety of estimates that can be produced from a survey, the standard deviation of an estimate is usually expressed relative to the estimate to which it pertains. This resulting measure, known as the coefficient of variation (CV) of an estimate, is obtained by dividing the standard deviation of the estimate by the estimate itself and is expressed as a percentage of the estimate.

For example, suppose hypothetically that it is estimated that 25% of Canadians aged 12 and over are regular smokers and that this estimate is found to have a standard deviation of 0.003. Then the CV of the estimate is calculated as:

$$(0.003/0.25) \times 100\% = 1.20\%$$

Statistics Canada commonly uses CV results when analyzing data and urges users producing estimates from the CCHS data files to also do so. For details on how to determine CVs, see Section 11. For guidelines on how to interpret CV results, see the table at the end of sub-section 10.4.

10. GUIDELINES FOR TABULATION, ANALYSIS AND RELEASE

This section of the documentation offers guidelines to users for tabulating, analyzing, publishing or otherwise releasing any estimates derived from the survey files. With the aid of these guidelines, users of microdata should be able to produce figures that are in close agreement with those produced by Statistics Canada. They will also be able to develop currently unpublished figures in a manner consistent with these established guidelines. Methods to measure precision and quality are also described along with release guidelines to help decide when an estimate should be used in publication.

10.1 Rounding guidelines

In order that estimates for publication or other release derived from the data files (Master, Share or PUMF) correspond to those produced by Statistics Canada, users are urged to adhere to the following guidelines regarding the rounding of such estimates:

a) Estimates in the main body of a statistical table are to be rounded to the nearest hundred units using the normal rounding technique. In normal rounding, if the first or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is raised by one. For example, in normal rounding to the nearest 100, if the last two digits are between 00 and 49, they are changed to 00 and the preceding digit (the hundreds digit) is left unchanged. If the last digits are between 50 and 99 they are changed to 00 and the proceeding digit is incremented by 1;

b) Marginal sub-totals and totals in statistical tables are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units using normal rounding[12];

c) Averages, proportions, rates and percentages are to be computed from unrounded components (i.e., numerators and/or denominators) and then are to be rounded themselves to one decimal using normal rounding. In normal rounding to a single digit, if the final or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is increased by 1;

d) Sums and differences of aggregates (or ratios) are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units (or the nearest one decimal) using normal rounding;

e) In instances where, due to technical or other limitations, a rounding technique other than normal rounding is used resulting in estimates to be published or otherwise released that differ from corresponding estimates published by Statistics Canada, users are urged to note the reason for such differences in the publication or release document(s);

f) Under no circumstances are unrounded estimates to be published or otherwise released by users. Unrounded estimates imply greater precision than actually exists.

10.2 Sample weighting guidelines for tabulation

The sample design used for this survey was not self-weighting. That is to say, the sampling weights are not identical for all individuals in the sample. When producing simple estimates, including the production of ordinary statistical tables, users must apply the proper sampling weight. If proper weights are not used, the estimates derived from the data file cannot be considered to be representative of the survey population, and will not correspond to those produced by Statistics Canada.

Users should also note that some software packages might not allow the generation of estimates that exactly match those available from Statistics Canada, because of their treatment of the weight field. If options are available, users should ensure that they specify that the weight is a sample weight rather than a frequency weight.

10.2.1 Definitions: categorical estimates, quantitative estimates

Before discussing how the survey data can be tabulated and analyzed, it is useful to describe the two main types of point estimates of population characteristics that can be generated from the data files.

Categorical estimates:

Categorical estimates are estimates of the number or percentage of the surveyed population possessing certain characteristics or falling into some defined category. The number of individuals who smoke daily is an example of such an estimate. An estimate of the number of persons possessing a certain characteristic may also be referred to as an estimate of an aggregate.

Example of categorical question:

*At the present do/does ... smoke cigarettes daily, occasionally or not at all? (**SMK_005**)*

__ Daily
__ Occasionally
__ Not at all

Quantitative estimates:

Quantitative estimates are estimates of totals or of means, medians and other measures of central tendency of quantities based upon some or all of the members of the surveyed population.

An example of a quantitative estimate is the average number of cigarettes smoked per day by individuals who smoke daily. The numerator is an estimate of the total number of cigarettes smoked per day by individuals who smoke daily, and its denominator is an estimate of the number of individuals who smoke daily.

Example of quantitative question:

*How many cigarettes do/does you/he/she smoke each day now? (**SMK_045**)*

|_|_| Number of cigarettes

10.2.2 Tabulation of categorical estimates

Estimates of the number of people with a certain characteristic can be obtained from the data file by summing the final weights of all records possessing the characteristic of interest.

Proportions and ratios of the form $\hat{X} / \hat{Y}$ are obtained by:

a) summing the final weights of records having the characteristic of interest for the numerator ($\hat{X}$);

b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$); then

c) dividing the numerator estimate by the denominator estimate.

10.2.3 Tabulation of quantitative estimates

Estimates of sums or averages for quantitative variables can be obtained using the following three steps (only step a) is necessary to obtain the estimate of a sum):

a) multiplying the value of the variable of interest by the final weight and summing this quantity over all records of interest to obtain the numerator ($\hat{X}$);
b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$); then
c) dividing the numerator estimate by the denominator estimate.

For example, to obtain the estimate of the average number of cigarettes smoked each day by individuals who smoke daily, first compute the numerator ($\hat{X}$) by summing the product between the value of variable **SMK_045** and the weight **WTS_M.** Next, sum this value over those records with a value of "daily" to the variable **SMK_005**. The denominator ($\hat{Y}$) is obtained by summing the final weight of those records with a value of "daily" to the variable **SMK_005**. Divide ($\hat{X}$) by ($\hat{Y}$) to obtain the average number of cigarettes smoked each day by daily smokers.

10.3 Guidelines for statistical analysis

The CCHS is based upon a complex design, with stratification and multiple stages of selection, and unequal probabilities of selection of respondents. Using data from such complex surveys presents problems to analysts because the survey design and the selection probabilities affect the estimation and variance calculation procedures that should be used. The use of the survey weights is the first step to ensuring that the proper results are obtained.

While many analysis procedures found in statistical packages allow weights to be used, the meaning or definition of the weight in these procedures can differ from what is appropriate in a

sample survey framework. The end result is that, while in many cases the estimates produced by the packages are correct, the variances and the statistics based on those variances are almost meaningless. To get the proper results, the weights must be correctly interpreted by the software package and the proper variance estimation techniques must be used.

To ensure that results from this complex design are meaningful, there are several options. The first and most appropriate option is to use the variance calculation options outlined in Chapter 11 based on the bootstrap methodology. Second, survey specific procedures exist in many packages that properly interpret the weight. For example, in SAS, PROC SURVEYMEANS would provide results that are more appropriate as the bootstrap weights are used for the variance estimates than the results produced from PROC MEANS. PROC MEANS can use the survey weights to correctly adjust estimates, taking into account the unequal weighting design, but it tends to underestimate the variances in not using the bootstrap weights and assuming the design is simple random sampling. Please consult the bootstrap variance estimation document found in the CCHS release package for more information. It is recommended that data users use software packages and functions that can incorporate both sample weights and bootstrap weights while performing their analyses.

Only the options outlined in Chapter 11 based on the bootstrap methodology will take the stratification, clustering and multiple frame design into account when calculating the variance.

10.4 Release guidelines

Since the estimates obtained from the survey are based on a sample, there is variability in the values obtained in the sense that a different sample could result in different results. To take this into consideration, users should first ensure that there are enough observations to properly estimate the statistic and also to estimate the variance. Once the variance is obtained, users should ensure that the variance is reasonable enough that the estimate can properly be interpreted as being near the true population value.

Master / Share

For users of the master or share files, it is recommended to have at least 10 observations with the characteristic of interest and 20 in the domain if a proportion is being calculated. With enough observations, the user can proceed to calculating the variance and the coefficients of variation using the bootstrap weights provided with the data along with the appropriate software to do the analysis. The variances can be used to calculate the CVs which aid in assessing the reliability of an estimate in regard to the sampling variability. The CV provides a relative measure of the sampling error as a proportion of the estimate. Estimates should be vetted using the guidelines in Table 10.1. CVs will increase as the variability of an estimate increases, and decrease as an estimate is more precise. However, an estimate can be precise while still having a large CV, specifically if the estimate has a value close to 0. Examining the confidence interval of the estimate will provide further indication of the quality of the estimate in terms of the variability. Long confidence intervals indicate less precision in the estimate while smaller confidence intervals indicate greater precision. When assessing the trustworthiness of sample proportions, the confidence intervals of estimates should be taken into account.

Table 10.1 Sampling variability guidelines

Type of Estimate	CV (in %)	Quality Indicator	Guidelines
Acceptable	CV $\leq$ 0.05	A	Estimates can be considered for general unrestricted release. Requires no special notation.
	0.05 < CV $\leq$ 0.15	B	
Marginal	0.15 < CV $\leq$ 0.25	C	Estimates can be considered for general unrestricted release but should be accompanied by a warning cautioning subsequent users of the high sampling variability associated with the estimates. Such estimates should be identified by the letter C or D.
	0.25 < CV $\leq$ 0.35	D	
Unacceptable	CV > 0.35	E	Statistics Canada recommends not to release estimates of unacceptable quality. However, if the user chooses to do so then estimates should be flagged with the letter E and the following warning should accompany the estimates: "The user is advised that . . .(specify the data) . . . do not meet Statistics Canada's quality standards for this statistical program. Conclusions based on these data will be unreliable and most likely invalid. These data and any consequent findings should not be published. If the user chooses to publish these data or findings, then this disclaimer must be published with the data."

11. APPROXIMATE SAMPLING VARIABILITY TABLES

For a quick approximation of coefficients of variation that will be applicable to a wide variety of categorical estimates produced from the Share file, a set of Approximate Sampling Variability

Tables will be produced. These "look-up" tables allow the user to obtain an approximate coefficient of variation based on the size of the estimate calculated from the survey data.

The coefficients of variation (CV) are derived using the variance formula for simple random sampling and incorporating a factor which reflects the multi-stage, clustered nature of the sample design. This factor, known as the ***design effect,*** was determined by first calculating design effects for a wide range of characteristics and then choosing, for each table produced, a conservative value among all design effects relative to that table. The value chosen was then used to generate a table that applies to the entire set of characteristics.

The Approximate Sampling Variability Tables, along with the design effects, the sample sizes and the population counts that were used to produce them, are provided in the document *Approximate Sampling Variability Tables*, which is available to the share file users. All coefficients of variation in the Approximate Sampling Variability Tables are approximate and, therefore, unofficial. Options concerning the computation of exact coefficients of variation are discussed in sub-section 11.7. Analysis using bootstrap weights is the method that should be employed for more precise results.

Remember: As indicated in Sampling Variability Guidelines in Section 10.4, if the number of observations on which an estimate is based is less than 10, the weighted estimate should not be released regardless of the value of the coefficient of variation. Coefficients of variation based on small sample sizes are too unpredictable to be adequately represented. Using the tables to estimate coefficients of variation are even less precise than using the bootstrap weights, so in this case, a minimum of 30 observations is required.

11.1 How to use the CV tables for categorical estimates

The following rules should enable the user to determine the approximate coefficients of variation from the Sampling Variability Tables for estimates of the number, proportion or percentage of the surveyed population possessing a certain characteristic and for ratios and differences between such estimates.

Rule 1: Estimates of numbers possessing a characteristic (aggregates)

The coefficient of variation depends only on the size of the estimate itself. On the appropriate Approximate Coefficients of Variations Table, locate the estimated number in the left-most column of the table (headed "Numerator of Percentage") and follow the asterisks (if any) across to the first figure encountered. Since not all the possible values for the estimate are available, the smallest value which is the closest must be taken (as an example, if the estimate is equal to 1,700 and the two closest available values are 1,000 and 2,000, the first has to be chosen). This figure is the approximate coefficient of variation.

Rule 2: Estimates of proportions or percentages of people possessing a characteristic

The coefficient of variation of an estimated proportion (or percentage) depends on both the size of the proportion and the size of the numerator upon which the proportion is based. Estimated proportions are relatively more reliable than the corresponding estimates of the numerator of the proportion when the proportion is based upon a sub-group of the population. This is due to the fact that the coefficients of variation of the latter type of estimates are based on the largest entry in a row of a particular table, whereas the coefficients of variation of the former type of estimators are based on some entry (not necessarily the largest) in that same row. (Note that in the tables the CVs decline in value reading across a row from left to right). For example, the estimated proportion of individuals who smoke daily out of those who smoke at all is more reliable than the estimated number who smoke daily.

When the proportion (or percentage) is based upon the total population covered by each specific table, the CV of the proportion is the same as the CV of the numerator of the proportion. In this case, this is equivalent to applying Rule 1.

When the proportion (or percentage) is based upon a subset of the total population (e.g., those who smoke at all), reference should be made to the proportion (across the top of the table) and to the numerator of the proportion (down the left side of the table). Since not all the possible values for the proportion are available, the smallest value which is the closest must be taken (for example, if the proportion is 23% and the two closest values available in the column are 20% and 25%, 20% must be chosen). The intersection of the appropriate row and column gives the coefficient of variation.

Rule 3: Estimates of differences between aggregates or percentages

The standard error of a difference between two estimates is approximately equal to the square root of the sum of squares of each standard error considered separately. That is, the standard error of a difference ($\hat{d} = \hat{X}_2 - \hat{X}_1$) is:

$$\sigma_{\hat{d}} = \sqrt{(\hat{X}_1 \alpha_1)^2 + (\hat{X}_2 \alpha_2)^2}$$

where $\hat{X}_1$ is estimate 1, $\hat{X}_2$ is estimate 2, and α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The coefficient of variation of $\hat{d}$ is given by $\sigma_{\hat{d}} / \hat{d}$. This formula is accurate for the difference between independent populations or subgroups, but is only approximate otherwise. It will tend to overstate the error, if $\hat{X}_1$ and $\hat{X}_2$ are positively correlated and understate the error if $\hat{X}_1$ and $\hat{X}_2$ are negatively correlated.

Rule 4: Estimates of ratios

In the case where the numerator is a subset of the denominator, the ratio should be converted to a percentage and Rule 2 applied. This would apply, for example, to the case where the denominator is the number of individuals who smoke at all and the numerator is the number of individuals who smoke daily out of those who smoke at all.

Consider the case where the numerator is not a subset of the denominator, as for example, the ratio of the number of individuals who smoke daily or occasionally as compared to the number of individuals who do not smoke at all. The standard deviation of the ratio of the estimates is approximately equal to the square root of the sum of squares of each coefficient of variation considered separately multiplied by $\hat{R}$, where $\hat{R}$ is the ratio of the estimates ($\hat{R} = \hat{X}_1 / \hat{X}_2$). That is, the standard error of a ratio is:

$$\sigma_{\hat{R}} = \hat{R}\sqrt{\alpha_1^2 + \alpha_2^2}$$

Where α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively.

The coefficient of variation of $\hat{R}$ is given by $\sigma_{\hat{R}} / \hat{R} = \sqrt{\alpha_1^2 + \alpha_2^2}$. The formula will tend to overstate the error, if $\hat{X}_1$ and $\hat{X}_2$ are positively correlated and understate the error if $\hat{X}_1$ and $\hat{X}_2$ are negatively correlated.

Rule 5: Estimates of differences of ratios

In this case, Rules 3 and 4 are combined. The CVs for the two ratios are first determined using Rule 4, and then the CV of their difference is found using Rule 3.

11.2 Examples of using the CV tables for categorical estimates

The following "real life" examples are included to assist users in applying the foregoing rules. Unrounded numbers are used in the examples to facilitate the step by step instructions. Users should still follow the rounding guidelines outlined in section 10.1 before publishing numbers.

Example 1: Estimates of numbers possessing a characteristic (aggregates)

Suppose that a user estimates that 3,665,449 individuals smoke daily in Canada. How does the user determine the coefficient of variation of this estimate?

1) Refer to the CANADA level CV table.

2) The estimated aggregate (3,665,449) does not appear in the left-hand column (the "Numerator of Percentage" column), so it is necessary to use the smallest figure closest to it, namely 3,000,000.

3) The coefficient of variation for an estimated aggregate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 2.3%.

4) So the approximate coefficient of variation of the estimate is 2.3%. According to the Sampling Variability Guidelines presented in Section 10.4, the finding that there were 3,665,449 individuals who smoke daily is publishable after applying the rounding rules.

Example 2: Estimates of proportions or percentages possessing a characteristic

Suppose that the user estimates that 3,665,449/5,151,237=71.2% of individuals in Canada who smoke at all smoke daily. How does the user determine the coefficient of variation of this estimate?

1) Refer to the CANADA level CV table.

2) Because the estimate is a percentage which is based on a subset of the total population (i.e., individuals who smoke at all, that is to say, daily or occasionally), it is necessary to use both the percentage (71.2%) and the numerator portion of the percentage (3,665,449) in determining the coefficient of variation.

3) The numerator (3,665,449) does not appear in the left-hand column (the "Numerator of Percentage" column) so it is necessary to use the smallest figure closest to it, namely 3,000,000. Similarly, the percentage estimate does not appear as any of the column headings, so it is necessary to use the smallest figure closest to it, 70.0%.

4) The figure at the intersection of the row and column used, namely 1.3% is the coefficient of variation (expressed as a percentage) to be used.

5) So the approximate coefficient of variation of the estimate is 1.3%. According to the Sampling Variability Guidelines presented in Section 10.4, the finding that 71.2% of individuals who smoke at all smoke daily can be published with no qualifications.

Example 3: Estimates of differences between aggregates or percentages

Suppose that a user estimates that, among men, 2,067955 /15,115,880 = 13.7% smoke daily (estimate 1), while for women, this percentage is estimated at 1,597,494 / 15,528,483 = 10.3% (estimate 2). How does the user determine the coefficient of variation of the difference between these two estimates?

1) Using the CANADA level CV table in the same manner as described in example 2 gives the CV for estimate 1 as 2.7% (expressed as a percentage), and the CV for estimate 2 as 3. 1% (expressed as a percentage).

2) Using rule 3, the standard error of a difference ($\hat{d} = \hat{X}_2 - \hat{X}_1$) is :

$$\sigma_{\hat{d}} = \sqrt{(\hat{X}_1 \alpha_1)^2 + (\hat{X}_2 \alpha_2)^2}$$

Where $\hat{X}_1$ is estimate 1, $\hat{X}_2$ is estimate 2, and α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The standard error of the difference $\hat{d}$ = (0.137 - 0.103) = 0.034 is :

$$\sigma_{\hat{d}} = \sqrt{[(0.137)(0.027)]^2 + [(0.103)(0.031)]^2}$$

$$= 0.00489$$

3) The coefficient of variation of $\hat{d}$ is given by $\sigma_{\hat{d}} / \hat{d}$ = 0.00489/0.034 = 0.144.

4) So the approximate coefficient of variation of the difference between the estimates is 14.6% (expressed as a percentage). According to the Sampling Variability Guidelines presented in Section 10.4, this estimate can be published.

Example 4: Estimates of ratios

Suppose that the user estimates that 3,665,449 individuals smoke daily, while 1,485,788 individuals smoke occasionally. The user is interested in comparing the estimate of daily to occasional smokers in the form of a ratio. How does the user determine the coefficient of variation of this estimate?

1) First of all, this estimate is a ratio estimate, where the numerator of the estimate (= $\hat{X}_1$) is the number of individuals who smoke occasionally. The denominator of the estimate (= $\hat{X}_2$) is the number of individuals who smoke daily.

2) Refer to the CANADA level CV table.

3) The numerator of this ratio estimate is 1,485,788. The smallest figure closest to it is 1,000,000. The coefficient of variation for this estimate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 4.0%.

4) The denominator of this ratio estimate is 3,665,449. The figure closest to it is 3,000,000. The coefficient of variation for this estimate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 2.3%.

5) So the approximate coefficient of variation of the ratio estimate is given by rule 4, which is,

$$\alpha_{\hat{R}} = \sqrt{\alpha_1^{\,2} + \alpha_2^{\,2}},$$

That is,

$$\alpha_{\hat{R}} = \sqrt{(0.040)^2 + (0.023)^2}$$

$$= 0.046$$

where α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The obtained ratio of occasional to daily smokers is 1,485,788/3,665,449 which is 0.41:1. The coefficient of variation of this estimate is 4.6% (expressed as a percentage), which is releasable with no qualifications, according to the Sampling Variability Guidelines presented in Section 10.4.

11.3 How to use the CV tables to obtain confidence limits

Although coefficients of variation are widely used, a more intuitively meaningful measure of sampling error is the confidence interval of an estimate. A confidence interval constitutes a statement on the level of confidence that the true value for the population lies within a specified range of values. For example a 95% confidence interval can be described as follows: if sampling of the population is repeated indefinitely, each sample leading to a new confidence interval for an estimate, then in 95% of the samples the interval will cover the true population value.

Using the standard error of an estimate, confidence intervals for estimates may be obtained under the assumption that under repeated sampling of the population, the various estimates obtained for a population characteristic are normally distributed about the true population value. Under this assumption, the chances are about 68 out of 100 that the difference between a sample estimate and the true population value would be less than one standard error, about 95 out of 100 that the

difference would be less than two standard errors, and about 99 out of 100 that the differences would be less than three standard errors. These different degrees of confidence are referred to as the confidence levels.

Confidence intervals for an estimate, $\hat{X}$, are generally expressed as two numbers, one below the estimate and one above the estimate, as $(\hat{X} - k, \hat{X} + k)$, where k is determined depending upon the level of confidence desired and the sampling error of the estimate.

Confidence intervals for an estimate can be calculated directly from the Approximate Sampling Variability Tables by first determining from the appropriate table the coefficient of variation of the estimate $\hat{X}$, and then using the following formula to convert to a confidence interval CI:

$$CI_X = [\ \hat{X} - z\ \hat{X}\ \alpha_{\hat{X}}\ , \hat{X} + z\ \hat{X}\ \alpha_{\hat{X}}\ ]$$

Where $\alpha_{\hat{X}}$ is determined coefficient of variation for $\hat{X}$, and

$z =$ 1 if a 68% confidence interval is desired

$z =$ 1.6 if a 90% confidence interval is desired

$z =$ 2 if a 95% confidence interval is desired

$z =$ 3 if a 99% confidence interval is desired.

Note: Release guidelines presented in section 10.4 which apply to the estimate also apply to the confidence interval. For example, if the estimate is not releasable, then the confidence interval is not releasable either.

11.4 Example of using the CV tables to obtain confidence limits

A 95% confidence interval for the estimated proportion of individuals who smoke daily from those who smoke at all (from example 2, sub-section 11.2) would be calculated as follows:

$\hat{X}$ = 0.712

z = 2

$\alpha_{\hat{X}}$ = 0.013 is the coefficient of variation of this estimate as determined from the tables.

$CI_{\hat{X}}$ -= {0.712 - (2) (0.712) (0.013), 0.712 + (2) (0.712) (0.013)}

$CI_{\hat{X}}$ -= {0.693, 0.731}

11.5 How to use the CV tables to do a Z-test

Standard errors may also be used to perform hypothesis testing, a procedure for distinguishing between population parameters using sample estimates. The sample estimates can be numbers, averages, percentages, ratios, etc. Tests may be performed at various levels of significance, where a level of significance is the probability of concluding that the characteristics are different when, in fact, they are identical.

Let $\hat{X}_1$ and $\hat{X}_2$ be sample estimates for 2 characteristics of interest. Let the standard error on the difference $\hat{X}_1 - \hat{X}_2$ be $\sigma_{\hat{d}}$. If the ratio of $\hat{X}_1 - \hat{X}_2$ over $\sigma_{\hat{d}}$ is between -2 and 2, then no conclusion about the difference between the characteristics is justified at the 5% level of significance. If however, this ratio is smaller than -2 or larger than +2, the observed difference is significant at the 5% level.

11.6 Example of using the CV tables to do a Z-test

Let us suppose we wish to test, at 5% level of significance, the hypothesis that there is no difference between the proportion of men who smoke daily AND the proportion of women who smoke daily. From example 3, sub-section 11.2, the standard error of the difference between these two estimates was found to be = 0.00489. Hence,

$$z = \frac{\hat{X}_1 - \hat{X}_2}{\sigma_{\hat{d}}} = \frac{0.137 - 0.103}{0.0048995} = \frac{0.034}{0.0048995} = 6.87$$

Since $z = 6.95$ is greater than 2, it must be concluded that there is a significant difference between the two estimates at the 5% level of significance. Note that the two sub-groups compared are considered as being independent, so the test is valid.

11.7 Exact variances/coefficients of variation

All coefficients of variation in the Approximate Sampling Variability Tables (CV Tables) are indeed approximate and, therefore, unofficial.

The computation of exact coefficients of variation is not a straightforward task since there is no simple mathematical formula that would account for all CCHS sampling frame and weighting aspects. Therefore, other methods such as resampling methods must be used in order to estimate measures of precision. Among these methods, the bootstrap method is the one recommended for analysis of CCHS data.

Many statistical packages allow for analyses using the bootstrap weights. More details are provided in section 10.3 and in the document on bootstrap variance estimation.

There are a number of reasons why a user may require an exact variance. A few are given below.

Firstly, if a user desires estimates at a geographic level other than those available in the tables (for example, at the rural/urban level), then the CV tables provided are not adequate. Coefficients of variation of these estimates may be obtained using "domain" estimation techniques through the exact variance program.

Secondly, should a user require more sophisticated analyses such as estimates of parameters from linear regressions or logistic regressions, the CV tables will not provide correct associated coefficients of variation. Although some standard statistical packages allow sampling weights to be incorporated in the analyses, the variances that are produced often do not take into account the stratified and clustered nature of the design properly, whereas the exact variance program would do so.

Thirdly, for estimates of quantitative variables, separate tables are required to determine their sampling error. Since most of the variables for the CCHS are primarily categorical in nature, this has not been done. Thus, users wishing to obtain coefficients of variation for quantitative variables can do so through the exact variance program. As a general rule, however, the coefficient of variation of a quantitative total will be larger than the coefficient of variation of the corresponding category estimate (i.e., the estimate of the number of persons contributing to the quantitative estimate). If the corresponding category estimate is not releasable, the quantitative estimate will not be either. For example, the coefficient of variation of the estimate of the total number of cigarettes smoked each day by individuals who smoke daily would be greater than the coefficient of variation of the corresponding estimate of the number of individuals who smoke daily. Hence if the coefficient of variation of the latter is not releasable, then the coefficient of variation of the corresponding quantitative estimate will also not be releasable.

Coefficients of variation produced by the tables are based on a wide range of variables and are therefore considered crude, whereas the exact variance program would give an exact coefficient of variation associated with the variable in question.

12. MICRODATA FILES: DESCRIPTION, ACCESS AND USE

The CCHS produces three types of microdata files: master files, share files and public use microdata files (PUMF). Table 12.1 includes the list of all available 2017 data files.

12.1 Master files

The master files contain all variables and all records from the survey collected during a collection period. These files are accessible at Statistics Canada for internal use and in Statistics Canada's Research Data Centres (RDC), and are also subject to custom tabulation requests.

For 2017, because collection only took place in half the communities for each territory, a single year of data is not representative of the territories. Because of this, the 2017 master file will only contain the data of respondents from the ten provinces. The data collected in 2017 for respondents from the territories will first be released on the 2017-2018 master data file.

12.1.1 Research Data Centre

The RDC Program enables researchers to use the survey data in the master files in a secure environment in several universities across Canada. Researchers must submit research proposals that, once approved, give them access to the RDC. For more information, please consult the following web page: https://www.statcan.gc.ca/eng/rdc/index

12.1.2 Custom tabulations

Another way to access the master files is to offer all users the option of having staff in Client Services of the Health Statistics Division prepare custom tabulations. This service is offered on a cost-recovery basis. It allows users who do not possess knowledge of tabulation software products to get custom results. The results are screened for confidentiality and reliability concerns before release. For more information, please contact Client Services at 613-951-1746 or by e-mail at statcan.hd-ds.statcan@canada.ca.

12.1.3 Remote access

Finally, the remote access service to the survey master files is another way to have access to these data if, for some reason, the user cannot access a Research Data Centre but have an approved research proposal. A researcher can be supplied with a synthetic or 'dummy' master file and a corresponding record layout. With these tools, the researcher can develop his or her own set of analytical computer programs. The code for the custom tabulations is then sent via e-mail to statcan.hd-ds.statcan@canada.ca. The code will then be transferred into Statistics Canada's internal secured network and processed using the appropriate master file of CCHS data. Estimates generated will be released to the user, subject to meeting the guidelines for analysis and release outlined in Section 10 of this document. Results are screened for confidentiality and reliability concerns and then the output is returned to the client. There is no charge for this service.

12.1.4 Real Time Remote Access (RTRA)

Users can access CCHS data using the Real Time Remote Access (RTRA) system, enabling fast access to Statistics Canada microdata. The RTRA system is an on-line remote access facility allowing users to run SAS programs, in real-time, against micro-data sets located in a central and secure location. Researchers using the RTRA system do not gain direct access to the micro-data and cannot view the content of the micro-data file. Instead, users submit SAS programs to extract results in the form of frequency tables. As RTRA researchers cannot view the micro-data, becoming a deemed employee of Statistics Canada is no longer necessary. This relationship is the basis that allows the RTRA to service its clients rapidly. Please see https://www.statcan.gc.ca/eng/rtra/rtra for more information.

12.2 Share files

The share files contain all variables and all records of CCHS respondents who agreed to share their data with Statistic Canada's partners, which are the provincial and territorial health departments, Health Canada and the Public Health Agency of Canada. Statistics Canada also asks respondents living in Quebec for their permission to share their data with the Institut de la statistique du Québec. The share file is released only to these organizations. Personal identifiers are removed from the share files to respect respondent confidentiality. Users of these files must first certify that they will not disclose, at any time, any information that might identify a survey respondent.

12.3 Public use microdata files

The public use microdata files (PUMF) are developed from the master files using a technique that balances the need to ensure respondent confidentiality with the need to produce the most useful data possible at the health region level. The PUMF must meet stringent security and confidentiality standards required by the *Statistics Act* before they are released for public access. To ensure that these standards have been achieved, each PUMF goes through a formal review and approval process by an executive committee of Statistics Canada.

Variables most likely to lead to identification of an individual are deleted from the data file or are collapsed to broader categories. Due to the risk of disclosure household weights are not included on the PUMF.

The PUMF contains the data collected over two years. It includes questions that were asked over two years. Unless otherwise specified, these questions are usually those included in the annual core content and in the two-year theme content as well as the optional content selected for two years by the provinces and territories.

There is no charge to access the PUMF in a post-secondary educational institution that is part of the Data Liberation Initiative, a partnership between post-secondary institutions and Statistics Canada for improving access to Canadian data resources. For more information on the Data Liberation Initiative please see http://www.statcan.gc.ca/eng/dli/dli.

The CCHS PUMF files are also free of charge to any researcher that contacts Client Services at 613-951-1746 or by e-mail at statcan.hd-ds.statcan@canada.ca.

Table 12.1 2017 CCHS data files

Reference Period	Files	File name	Sampling weight	Bootstrap weights file	Variables included	Records included
2017	Main master file	HS.txt	WTS_M	bsw.txt	All common and all optional modules.	All respondent records in the ten provinces
	Share file	HS.txt	WTS_S	bsw.txt	All common and all optional modules.	Records of all respondents who agreed to share their data
2017	Rapid response file	HS_PCN.txt	WTS_M	bsw.txt	All core modules and questions in the PCN rapid response.	All respondents in the ten provinces selected over a six month collection period (January to June 2017)

12.4 How to use the CCHS data files: annual data file or two-year data file?

Users who have access to share files or master files have the choice of using one-year or two-year data files. Decisions about which period to use in a given data analysis should be guided by the level of detail and the quality required. With a one-year file, estimates will not always be available because of the quality associated with limited sample sizes. Since the territories are not available on the one-year file, users wanting true national estimates should use the two-year file.

Before interpreting and using a CCHS estimate, it is recommended to make sure that the estimates meets the following rules:

- Coefficient of Variation 35.0% or less
- a minimum of 10 respondents in the domain with the characteristic and
- total domain of interest includes at least 20 respondents (for proportions or ratios)

This will not be possible for rare characteristics and detailed domains with one-year files. Instead, users will have to rely on two-year files or multi-year files. For more information on combining CCHS cycles into a multi-year file, please see https://www150.statcan.gc.ca/n1/pub/82-003-x/2009001/article/10795-eng.pdf

Where the use of either a one-year or two-year file is viable, the user should consider the trade-off between accuracy and timeliness. If it is important to reflect the current characteristics of a population as closely as possible, the one-year file would be preferable. However, with the increased sample size, more detailed estimates and analyses can be carried out with a two-year file.

12.5 Use of weight variable

The weight variable **WTS_M** represents the sampling weight for key survey files. For a given respondent, the sampling weight can be interpreted as the number of people the respondent represents in the Canadian population. This weight must always be used when computing statistical estimates in order to make inference at the population level possible. The production of unweighted estimates is not recommended. The sample allocation, as well as the survey design specifics can cause such results to not correctly represent the population. Refer to section 8 on weighting for a more detailed explanation on the creation of this weight. The weight variable **WTS_M** must be used for regional analyses.

The Food security (FSC) module, included in certain reference period data files, measures concepts that apply not only to the respondent's situation, but also to that of the respondent's entire household. Depending on the level of analysis, the analysis of the variables may require use of a weight calculated to represent the number of Canadian households, rather than the number of persons. This weight variable **WTS_HH** is found in a separate file (HS_HHWT.txt). It can be used in place of the variable **WTS_M** for household analyses at the national and provincial levels.

12.6 Variable naming convention

The variable naming convention adopted allows data users to easily use and identify the data based on the module and variable type. The CCHS variable naming convention fulfils two requirements: to restrict variable names to a maximum of eight characters for ease of use by analytical software products and to identify easily conceptually identical variables from one survey collection period to the next. Questions to which changes are made between two collection periods, and where the changes alter the concept measured by the question, are entirely renamed to avoid any confusion in the analysis.

A variable (REFPER, format = YYYYMM-YYYYMM) was added to the microdata files in order to identify the beginning and the end of the reference during which data included in the file were collected. This variable will be useful, notably for users wanting to use data from several collection periods at a time. Therefore, variable names for identical modules or questions from one collection year to the next (e.g., 2015 and 2017) will be the same. However, some minor changes have been made to variable names since the 2015 redesign.

The naming convention used for variables beginning with the 2015 CCHS use up to eight characters. The variable names are structured as follows:

Positions 1 to 3:	Module/questionnaire section name
Position 4:	Variable type (underscore, C,D, F or G)
Positions 5 to 8:	Question number and answer option for multiple response questions in increments of five (e.g. 005, 010, 015)

Example 1 shows that the structure of the variable name for question 005, Smoking Module, is SMK_005 :

Positions 1 to 3:	SMK	Smoking module
Position 4 :	_	(underscore = collected data)
Position 5 to 8:	005	Question number

Example 2 shows the structure of the variable name for question 2 of the Contacts with health professionals – part 1 Module (CHP_015), which is a multi-response question:

Positions 1 to 3:	CHP	Health care utilization module
Position 4 :	_	(underscore = collected data)
Position 5 to 8:	015	Corresponding question number and answer option

Positions 1 to 3 contain the acronyms for each of the modules. These acronyms appear beside the module names given in the table in Appendix A. It should be noted that some module acronyms consist of four characters. Variables names from these modules are still bound by a maximum of 8 characters.

Position 4 designates the variable type based on whether it is a variable collected directly from a questionnaire question (“_”), derived (“D”), flag (“F”), or grouped (“G”) variable.

In general, the last four positions (5 to 8) follow the variable numbering used on the questionnaire. The letter "Q" used to represent the word "question" is removed, and all question numbers are presented in a two or three digit format. For example, question Q005A in the questionnaire becomes simply 005A, and question Q010 becomes simply 010.

Table 12.2 Designation of codes used in the 4th position of the CCHS variable names

_	Collected variable	A variable that appears directly on the questionnaire
C	Coded variable	A variable coded from one or more collected variables (e.g., SIC, Standard Industrial Classification code)
D	Derived variable	A variable calculated from one or more collected or coded variables, usually calculated during head office processing (e.g., Health Utility Index)
F	Flag variable	A variable calculated from one or more collected variables (like a derived variable), but usually calculated by the data collection computer application for later use during the interview (e.g., work flag)
G	Grouped variable	Collected, coded, suppressed or derived variables collapsed into groups (e.g., age groups)

For questions that have more than one response option, the final position in the variable naming sequence is represented by a letter. For this type of question, new variables were created to differentiate between a "yes" or "no" answer for each response option. For example, if Q010 had 4 response options, the new questions would be named Q010A for option 1, Q010B for option 2, Q010C for option 3, and Q010D for option 4. If only options 2 and 3 were selected, then Q010A = No, Q010B = Yes, Q010C = Yes and Q010D = No.

12.7 Data dictionaries

Separate data dictionary reports, including variable names, concepts, universe statements and frequencies, are provided for the main master and share files and each of rapid response files (if applicable). All variables on these files will appear in the data dictionary report, though some may not include a frequency (SampleID for example).

When a variable includes the frequencies for responses, the report will show four columns:

1. Content – The text label for each of the response categories.
2. Code – The numerical value associated with each category.
3. Sample – The unweighted frequency on the file of respondents with each response.
4. Population – The weighted frequency for the response.

In the master file data dictionary reports, optional content modules are treated in the same way as previous CCHS cycles. For each module, a flag indicates whether a given respondent lives in a health region where the module was selected as optional content. When the flag is equal to 2 (No), all variables in the module have “not applicable” values. For example, the DOWST variable indicates if the Work stress module applies to a given respondent.

12.8 Differences in calculation of core and theme content variables using different files

Variables from core content modules can be estimated using either of the two data files provided, when a one year and a two-year data file is available. Depending on which file is used, very small differences will be observed.

All official Statistics Canada estimates of variables from common modules are based on the main master file sampling weight.

Appendix A – Canadian community health survey content overview (2017-2018)

Core content (provinces)

- Alcohol use (ALC)
- Chronic conditions (CCC)
- Flu shots (FLU)
- General health (GEN)
- Height and weight – self-reported (HWT)
- Maternal experiences (MEX)
- Physical activities – adults 18 years and older (PAA)
- Physical activities for youth (PAY)
- Primary health care (PHC)
- Smoking (SMK)

Administration and socio-demographics

- Administration information (ADM)
- Age of respondent (ANC1)
- Income (INC)
- Labour force (LBF)
- Main activity (MAC)
- Person most knowledgeable about household situation (PMK)
- Socio-demographic characteristics (SDC)

Two-year theme content (provinces)

- Chronic conditions (CCC – theme 2)[13]
- Food security (FSC)
- Injuries (INJ)
- Alcohol use during maternal experience (MXA)
- Smoking during maternal experience (MXS)
- Sedentary behaviours (SBE)
- Washington Group – disability measure (WDM)

[13] In 2017 and 2018, CCC core content was asked in the provinces and territories. In 2017, theme 2 content was asked in the provinces and territories. In 2018, a shorter version of theme 2 was asked in the provinces, while a regular version of theme 2 was asked in the territories. In 2018, one new question (CCC_087) was asked only in the provinces. For details, please see the CCC table at the end of Appendix A.

One-year theme content (provinces)

2017	2018
• Colorectal cancer testing (CCT) • Fruit and vegetable consumption (FVC) • Mammography (MAM) • Pap smear test (PAP) • Use of protective equipment (UPE)	• Dental care (DEN) • Home care services – with palliative care (HMC) • Prescriptions – cost-related non-adherence (short version) (PCN2) • Perceived need for care (PNC) • Pain relief medication (PRM) • Stress and relationships (STR)

Rapid response (provinces)

2017	2018
Barriers to physical activity (BPA) July – December	Antibiotic medication use (AMU) January – June Gambling (GAM) July – December

Core content (territories)

- Alcohol use (ALC)
- Chronic conditions (CCC)
- Flu shots (FLU)
- General health (GEN)
- Height and weight – self-reported (HWT)
- Maternal experiences (MEX)

- Physical activities – adults 18 years and older (PAA)
- Physical activities for youth (PAY)
- Primary health care (PHC)
- Smoking (SMK)
- Fruit and vegetable consumption (FVC)

Administration and socio-demographics

- Administration information (ADM)
- Age of respondent (ANC1)
- Income (INC)
- Labour force (LBF)
- Main activity (MAC)
- Person most knowledgeable about household situation (PMK)
- Socio-demographic characteristics (SDC)

Two-year theme content (territories)[14]

- Chronic conditions (CCC – theme 2)[15]
- Food security (FSC)
- Injuries (INJ)
- Alcohol use during maternal experience (MXA)
- Smoking during maternal experience (MXS)
- Sedentary behaviours (SBE)
- Washington Group – disability measure (WDM)
- Colorectal cancer testing (CCT)
- Mammography (MAM)
- Pap smear test (PAP)
- Use of protective equipment (UPE)

[14] There is no one-year theme content for the territories because a module must be asked for both 2017 and 2018 in order to be disseminated for a territory.

[15] In 2017 and 2018, CCC core content was asked in the provinces and territories. In 2017, theme 2 content was asked in the provinces and territories. In 2018, a shorter version of theme 2 was asked in the provinces, while a regular version of theme 2 was asked in the territories. In 2018, one new question (CCC_087) was asked only in the provinces. For details, please see the CCC table at the end of Appendix A.

Appendix A – Canadian Community Health Survey. Chronic conditions (CCC) – 2017-2018

Content type and year		Provinces		Territories
Core content	Y	Core content – 2017-2018	Y	Core content – 2017-2018
2017 – theme 2 content	Y	T2 content	Y	T2 content
2018 – theme 2 content	Y	T2 content (short version) – excludes questions CCC_150, CCC_170, CCC_175, CCC_180	Y	T2 content (regular version) – includes questions CCC_150, CCC_170, CCC_175, CCC_180
New question (CCC_087) for 2018	Y	Question CCC_087 – asked in 2018 only	N	Question CCC_087 – not asked in 2018

Chronic conditions 2017-2018		Provinces			Territories		
Question	**Concept**	**2017**	**2018**	**2017-2018**	**2017**	**2018**	**2017-2018**
CCC_005	Joint pain – 30 d	N	N	N	N	N	N
CCC_010	Joint pain symptoms – 3 mo	N	N	N	N	N	N
CCC_015	Has asthma	Y	Y	Y	Y	Y	Y
CCC_020	Asthma – had symptoms / attacks – 12 mo	Y	Y	Y	Y	Y	Y
CCC_025	Asthma – took medication – 12 mo	Y	Y	Y	Y	Y	Y
CCC_030	Has a COPD	Y	Y	Y	Y	Y	Y
CCC_035	Has sleep apnea	N	N	N	N	N	N
CCC_040	Has scoliosis	N	N	N	N	N	N
CCC_045	Has fibromyalgia	N	N	N	N	N	N
CCC_050	Has arthritis (e.g. osteoarthritis, rheumatoid arthritis, gout)	Y	Y	Y	Y	Y	Y
CCC_055	Has back problems (excluding scoliosis, fibromyalgia and arthritis)	N	N	N	N	N	N
CCC_060	Has osteoporosis	N	N	N	N	N	N
CCC_065	Has high blood pressure	Y	Y	Y	Y	Y	Y
CCC_070	High blood pressure – took medication – 1 mo	Y	Y	Y	Y	Y	Y
CCC_075	Has high blood cholesterol / lipids	Y	Y	Y	Y	Y	Y
CCC_080	High blood cholesterol / lipids - took medication – 1 mo	Y	Y	Y	Y	Y	Y
CCC_085	Has heart disease	Y	Y	Y	Y	Y	Y
CCC_087	Heart disease–ever diagnosed	N	Y	N	N	N	N
CCC_090	Suffers from the effects of a stroke	Y	Y	Y	Y	Y	Y
CCC_095	Has diabetes	Y	Y	Y	Y	Y	Y
CCC_100	Diabetes – age first diagnosed	Y	Y	Y	Y	Y	Y

Content type and year

Core content: 2015-16 and 2017-18
Theme 2: 2017-2018
Theme 1: 2015-2016

Chronic conditions 2017-2018		Provinces			Territories		
Question	**Concept**	**2017**	**2018**	**2017-2018**	**2017**	**2018**	**2017-2018**
CCC_105	Diabetes – diagnosed – during pregnancy	Y	Y	Y	Y	Y	Y
CCC_110	Diabetes – diagnosed – other than during pregnancy	Y	Y	Y	Y	Y	Y
CCC_115	Diabetes – time between diagnosis and start of insulin	Y	Y	Y	Y	Y	Y
CCC_120	Diabetes – currently takes insulin	Y	Y	Y	Y	Y	Y
CCC_125	Diabetes – took pills to control blood sugar – 1 mo	Y	Y	Y	Y	Y	Y
CCC_130	Has cancer	Y	Y	Y	Y	Y	Y
CCC_135	Has been diagnosed with cancer – lifetime	Y	Y	Y	Y	Y	Y
CCC_140	Has migraine headaches	N	N	N	N	N	N
CCC_145	Has Alzheimer's Disease or any other dementia	Y	Y	Y	Y	Y	Y
CCC_150	Has intestinal or stomach ulcers	Y	N	N	Y	Y	Y
CCC_155	Has a bowel disorder (Crohn's, ulcerative colitis, IBS, incontinence)	Y	Y	Y	Y	Y	Y
CCC_160	Type of bowel disease	Y	Y	Y	Y	Y	Y
CCC_165	Has urinary incontinence	Y	Y	Y	Y	Y	Y
CCC_170	Has allergies – was told by a health professional	Y	N	N	Y	Y	Y
CCC_175	Source(s) of allergy	Y	N	N	Y	Y	Y
CCC_180	Had hay fever or nasal allergy – lifetime	Y	N	N	Y	Y	Y
CCC_185	Has chronic fatigue syndrome	N	N	N	N	N	N
CCC_190	Suffers from multiple chemical sensitivities	N	N	N	N	N	N
CCC_195	Has a mood disorder (depression, bipolar, mania, dysthymia)	Y	Y	Y	Y	Y	Y
CCC_200	Has an anxiety disorder (phobia, OCD, panic)	Y	Y	Y	Y	Y	Y

Content type and year

Core content: 2015-16 and 2017-18
Theme 2: 2017-2018
Theme 1: 2015-2016

Appendix B – Selection of optional content by province and territory (2017)

Appendix B – Optional content selection by province and territory (2017)*

Description	N.L.	P.E.I.	N.S.	N.B.	Que.	Ont.	Man.	Sask.	Alta.	B.C.	Y.T.*	N.W.T.*	Nvt*
ACC Access to health care system	☐	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑
ADL Activities of daily living	☐	☑	☐	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐
ALW Alcohol use during the past week	☐	☑	☐	☐	☐	☑	☐	☐	☐	☑	☐	☐	☐
BPC Blood pressure check	☑	☐	☑	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐
FGU Canada's Food Guide use	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐
CIH Changes made to improve health	☑	☑	☐	☑	☐	☐	☐	☐	☐	☑	☐	☐	☐
CMH Consultations about mental health	☑	☐	☑	☐	☑	☐	☐	☐	☐	☐	☐	☐	☑
SPC Consultations with health professionals on smoking	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐
CP2 Contacts with health professionals - extended block	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐
DEN Dental care	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☑	☐	☑
DEP Depression	☐	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐
DIA Diabetes care	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
DIS Distress	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☑	☐
DRV Driving and safety	☐	☐	☐	☐	☐	☑	☐	☐	☑	☐	☐	☐	☐
DRG Drug use	☐	☐	☐	☑	☐	☑	☑	☐	☐	☐	☑	☑	☐
DRM Drug use methods	☐	☐	☐	☑	☐	☐	☑	☐	☐	☐	☐	☐	☐
ETS Exposure to second hand smoke	☐	☑	☐	☑	☑	☐	☐	☐	☐	☐	☐	☐	☐
FDC Food Choices	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐
INS Health insurance coverage	☐	☐	☐	☑	☐	☐	☑	☐	☐	☐	☐	☐	☐

Note: Although they are collected in the CCHS every year, respondents living in the territories are only included on two year microdata files.

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Description	N.L.	P.E.I.	N.S.	N.B.	Que.	Ont.	Man.	Sask.	Alta.	B.C.	Y.T.*	N.W.T.*	Nvt*
HMC Home care services	☑	☐	☑	☑	☐	☐	☑	☐	☐	☐	☐	☐	☐
MDA Medical doctor attachment	☐	☐	☐	☑	☐	☐	☑	☐	☐	☑	☐	☐	☐
MED Medication use	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐
OHT Oral health	☐	☐	☐	☐	☑	☑	☐	☐	☐	☐	☐	☐	☑
PSC Patient satisfaction - community-based care	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
PSA Prostate cancer screening	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☑	☐	☐
SWL Satisfaction with life	☐	☐	☑	☐	☑	☐	☐	☐	☑	☐	☑	☐	☐
SLP Sleep	☐	☑	☐	☐	☑	☐	☐	☐	☑	☑	☑	☐	☑
SCA Smoking cessation methods	☐	☐	☑	☐	☐	☐	☐	☐	☑	☐	☑	☐	☑
SPS Social provisions (social support)	☑	☑	☐	☐	☐	☐	☐	☐	☑	☑	☐	☐	☐
STS Sources of stress	☐	☐	☐	☐	☑	☐	☐	☐	☑	☑	☐	☑	☐
SSB Sun safety behaviors	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐	☐	☐	☐
TAL Tobacco products - alternatives	☐	☐	☐	☐	☑	☑	☐	☐	☐	☐	☑	☐	☐
UCN Unmet care needs	☐	☐	☑	☑	☐	☑	☑	☐	☑	☐	☐	☐	☐
INW Workplace injury	☐	☐	☐	☐	☐	☐	☐	☐	☐	☑	☐	☐	☐

Note: Although they are collected in the CCHS every year, respondents living in the territories are only included on two year microdata files.

Appendix C - Available geography in the master and share files and their corresponding codes: Canada, provinces, health regions and peer groups

Appendix C - Available geography in the master and share files and their corresponding codes: Canada, provinces, health regions and peer groups

0	Canada	
10	Newfoundland and Labrador	
1011-C		Eastern Regional Integrated Health Authority
1012-E		Central Regional Integrated Health Authority
1013-E		Western Regional Integrated Health Authority
1014-E		Labrador-Grenfell Regional Integrated Health Authority
11	Prince Edward Island	
1100-C		Prince Edward Island
12	Nova Scotia	
1201-E		Zone 1 - Western
1202-C		Zone 2 - Northern
1203-E		Zone 3 - Eastern
1204-A		Zone 4 - Central
13	New Brunswick	
1301-C		Zone 1 (Moncton area)
1302-A		Zone 2 (Saint John area)
1303-C		Zone 3 (Fredericton area)
1304-E		Zone 4 (Edmundston area)
1305-E		Zone 5 (Campbellton area)
1306-E		Zone 6 (Bathurst area)
1307-E		Zone 7 (Miramichi area)
24	Quebec	
2401-C		Région du Bas–Saint–Laurent
2402-C		Région du Saguenay — Lac–Saint–Jean
2403-A		Région de la Capitale–Nationale
2404-C		Région de la Mauricie et du Centre–du–Québec
2405-C		Région de l'Estrie
2406-G		Région de Montréal
2407-B		Région de l'Outaouais
2408-C		Région de l'Abitibi–Témiscamingue
2409-C		Région de la Côte–Nord
2410-C		Région du Nord–du–Québec
2411-E		Région de la Gaspésie — Îles–de–la–Madeleine
2412-D		Région de la Chaudière–Appalaches
2413-A		Région de Laval

2414-B		Région de Lanaudière
2415-B		Région des Laurentides
2416-B		Région de la Montérégie
35	Ontario by Local Health Integration Network	
3501		Erie St. Clair
3502		South West
3503		Waterloo Wellington
3504		Hamilton Niagara Haldimand Brant
3505		Central West
3506		Mississauga Halton
3507		Toronto Central
3508		Central
3509		Central East
3510		South East
3511		Champlain
3512		North Simcoe Muskoka
3513		North East
3514		North West
35	Ontario by Health Unit	
3526-E		The District of Algoma Health Unit
3527-A		Brant County Health Unit
3530-B		Durham Regional Health Unit
3531-C		Elgin-St Thomas Health Unit
3533-D		Grey Bruce Health Unit
3534-D		Haldimand-Norfolk Health Unit
3535-D		Haliburton, Kawartha, Pine Ridge District Health Unit
3536-B		Halton Regional Health Unit
3537-A		City of Hamilton Health Unit
3538-C		Hastings and Prince Edward Counties Health Unit
3539-D		Huron County Health Unit
3540-E		Chatham-Kent Health Unit
3541-C		Kingston, Frontenac and Lennox and Addington Health Unit
3542-C		Lambton Health Unit
3543-D		Leeds, Grenville and Lanark District Health Unit
3544-A		Middlesex–London Health Unit
3546-A		Niagara Regional Area Health Unit
3547-C		North Bay Parry Sound District Health Unit

3549-C		Northwestern Health Unit
3551-B		City of Ottawa Health Unit
3552-D		Oxford County Health Unit
3553-H		Peel Regional Health Unit
3554-D		Perth District Health Unit
3555-C		Peterborough County–City Health Unit
3556-C		Porcupine Health Unit
3557-D		Renfrew County and District Health Unit
3558-D		The Eastern Ontario Health Unit
3560-B		Simcoe Muskoka District Health Unit
3561-C		Sudbury and District Health Unit
3562-C		Thunder Bay District Health Unit
3563-C		Timiskaming Health Unit
3565-B		Waterloo Health Unit
3566-B		Wellington–Dufferin-Guelph Health Unit
3568-A		Windsor–Essex County Health Unit
3570-H		York Regional Health Unit
3595-G		City of Toronto Health Unit
46	Manitoba	
4601-A		Winnipeg Regional Health Authority
4602-D		Prairie Mountain Health
4603-D		Interlake-Eastern Regional Health
4604-F		Northern Regional Health Authority
4605-B		Southern Health
47	Saskatchewan	
4701-D		Sun Country Regional Health Authority
4702-D		Five Hills Regional Health Authority
4703-D		Cypress Regional Health Authority
4704-B		Regina Qu'Appelle Regional Health Authority
4705-D		Sunrise Regional Health Authority
4706-B		Saskatoon Regional Health Authority
4707-D		Heartland Regional Health Authority
4708-C		Kelsey Trail Regional Health Authority
4709-C		Prince Albert Parkland Regional Health Authority
4710-B		Prairie North Regional Health Authority
4714-F		Mamawetan/Keewatin/Athabasca Regional Health Authorities[16]

16. For most data sources (with the exception of Census and Demographic population estimates), health region level data are not available for some northern health regions in Saskatchewan which have small populations. To avoid suppression in these areas where small numbers or sample size impact on data quality, data have been grouped with

48	Alberta	
4831-B		South Zone
4832-B		Calgary Zone
4833-B		Central Zone
4834-B		Edmonton Zone
4835-I		North Zone
59	British Columbia	
5911-D		East Kootenay Health Service Delivery Area
5912-C		Kootenay-Boundary Health Service Delivery Area
5913-A		Okanagan Health Service Delivery Area
5914-C		Thompson/Cariboo Health Service Delivery Area
5921-B		Fraser East Health Service Delivery Area
5922-H		Fraser North Health Service Delivery Area
5923-H		Fraser South Health Service Delivery Area
5931-H		Richmond Health Service Delivery Area
5932-G		Vancouver Health Service Delivery Area
5933-A		North Shore/Coast Garibaldi Health Service Delivery Area
5941-A		South Vancouver Island Health Service Delivery Area
5942-C		Central Vancouver Island Health Service Delivery Area
5943-C		North Vancouver Island Health Service Delivery Area
5951-C		Northwest Health Service Delivery Area
5952-C		Northern Interior Health Service Delivery Area
5953-I		Northeast Health Service Delivery Area
A	Peer group A	
B	Peer group B	
C	Peer group C	
D	Peer group D	
E	Peer group E	
F	Peer group F	
G	Peer group G	
H	Peer group H	

neighbouring regions, as follows: Athabasca Health Authority, Saskatchewan (4713) is combined with Mamawetan Churchill River Regional Health Authority (4711) and Keewatin Yatthé Regional Health Authority (4712) and referred to as "Mamawetan/Keewatin/Athabasca Regional Health Authorities (4714)"

I	Peer group I

Appendix D (2017) Sample allocation by health region and age group and sample allocation by Local Health Integrated Network (LHIN) and age group in Ontario

Appendix D (2017) Sample allocation by health region and age group

		Adults		Children		Combined	
Prov./Terr.	**Health Region**	**expected # of respondents**	**raw sample size**	**expected # of respondents**	**raw sample size**	**expected # of respondents**	**raw sample size**
Canada	**Total**	59,448	122,513	5,042	8,605	64,490	131,117
N.L.	**Total**	1,586	3,089	157	273	1,743	3,362
	1011	581	1,233	56	100	636	1,333
	1012	378	731	38	68	416	800
	1013	359	605	35	60	394	665
	1014	269	519	28	45	297	564
P.E.I.	**Total**	1,058	2,357	111	203	1,168	2,560
	1100	1,058	2,357	111	203	1,168	2,560
N.S.	**Total**	2,486	4,926	193	319	2,678	5,244
	1201	599	1,156	47	82	646	1,239
	1202	541	1,036	43	78	584	1,114
	1203	556	1,231	43	71	599	1,302
	1204	789	1,503	60	87	849	1,590
N.B.	**Total**	2,091	4,829	164	301	2,255	5,131
	1301	400	968	30	55	430	1,023
	1302	375	834	31	50	406	884
	1303	370	741	30	37	400	779
	1304	241	618	18	40	259	658
	1305	192	451	15	29	207	480
	1306	284	709	22	55	306	764
	1307	229	507	19	35	247	543
Que.	**Total**	12,402	23,697	948	1,368	13,350	25,064
	2401	600	1,129	46	68	646	1,198
	2402	718	1,327	52	64	770	1,392
	2403	915	1,645	70	100	985	1,745
	2404	816	1,402	65	98	881	1,500
	2405	738	1,400	56	78	794	1,478
	2406	1,481	3,205	101	180	1,582	3,384
	2407	761	1,618	60	78	821	1,696
	2408	600	1,155	44	63	644	1,218
	2409	550	1,111	35	51	585	1,163
	2410	244	470	22	30	266	500
	2411	550	1,086	35	56	585	1,142
	2412	776	1,315	61	75	837	1,390
	2413	777	1,500	64	101	841	1,601
	2414	805	1,500	66	88	871	1,588
	2415	843	1,672	71	103	914	1,775
	2416	1,228	2,160	98	135	1,326	2,295
Ont.	**Total**	18,177	37,740	1,516	2,552	19,693	40,292
	3526	382	1,020	30	48	412	1,068

	3527	398	709	34	68	432	777
	3530	689	1,444	61	101	750	1,545
	3531	346	606	31	60	376	666
	3533	431	964	35	60	466	1,024
	3534	375	794	32	52	407	846
	3535	449	910	36	50	485	960
	3536	642	1,236	56	94	697	1,330
	3537	655	1,479	55	98	710	1,577
	3538	427	954	34	60	461	1,014
	3539	302	575	26	38	328	613
	3540	371	653	32	52	402	706
	3541	460	1,057	36	52	496	1,109
	3542	394	764	33	43	427	807
	3543	437	858	36	65	473	923
	3544	620	1,212	51	112	671	1,324
	3546	615	1,314	50	77	665	1,391
	3547	394	830	31	41	425	871
	3549	310	727	27	48	336	775
	3551	787	1,478	65	112	852	1,590
	3552	370	670	33	57	403	727
	3553	902	1,941	80	131	982	2,072
	3554	327	538	28	43	356	580
	3555	411	812	32	48	443	860
	3556	332	629	28	50	361	678
	3557	362	799	29	49	391	848
	3558	459	882	39	58	498	940
	3560	649	1,549	54	84	703	1,634
	3561	458	971	37	69	495	1,040
	3562	418	898	34	55	452	953
	3563	249	481	20	31	269	512
	3565	649	1,324	55	95	704	1,419
	3566	518	1,038	45	58	562	1,096
	3568	587	1,270	50	109	637	1,379
	3570	830	1,781	73	126	903	1,907
	3595	1,172	2,576	91	157	1,263	2,734
Man.	**Total**	2,902	5,469	261	455	3,163	5,924
	4601	964	1,714	82	120	1,046	1,835
	4602	553	1,012	49	94	602	1,105
	4603	500	1,076	44	72	544	1,148
	4604	316	688	30	86	346	774
	4605	569	979	56	82	624	1,062
Sask.	**Total**	2,564	5,133	226	408	2,790	5,541
	4701	213	384	19	26	232	410
	4702	212	392	19	35	231	427

	4703	198	396	17	23	215	419
	4704	374	677	32	50	406	727
	4705	211	438	19	24	230	462
	4706	405	784	35	57	440	841
	4707	196	385	18	19	214	404
	4708	186	407	16	37	202	444
	4709	230	478	21	51	251	529
	4710	220	446	20	41	240	487
	4714	119	345	11	44	130	389
Alta.	**Total**	6,992	15,211	588	1,006	7,580	16,217
	4831	1,023	2,188	89	150	1,112	2,338
	4832	1,846	4,095	150	232	1,996	4,327
	4833	1,213	2,484	105	166	1,317	2,650
	4834	1,740	3,608	142	247	1,882	3,856
	4835	1,171	2,835	102	211	1,273	3,046
B.C.	**Total**	8,194	18,180	629	1,223	8,823	19,403
	5911	350	773	27	53	376	826
	5912	352	717	27	54	379	771
	5913	583	1,229	44	86	627	1,315
	5914	496	1,184	38	62	534	1,246
	5921	537	1,235	44	84	581	1,319
	5922	722	1,601	55	121	777	1,722
	5923	757	1,632	62	123	819	1,755
	5931	485	1,075	37	72	522	1,147
	5932	755	1,892	52	95	806	1,987
	5933	546	1,264	42	83	588	1,348
	5941	604	1,209	44	92	648	1,301
	5942	535	1,179	39	72	574	1,251
	5943	402	821	31	54	433	874
	5951	320	736	26	42	346	777
	5952	424	960	34	66	458	1,026
	5953	329	673	27	64	355	737
Y.T.	**6001**	385	732	83	156	468	888
N.W.T.	**6101**	382	737	83	170	465	907
Nvt.	**6201**	231	413	83	171	314	584

Appendix D (2017) - Sample allocation by Local Health Integrated Network (LHIN) and frame in Ontario

		Adults	
Prov./Terr.	**LHIN**	**expected # of respondents**	**raw sample size**
Ont.	**Total**	18,177	37,740
	3501	1,352	2,687
	3502	2,396	4,564
	3503	1,056	2,123
	3504	2,283	4,755
	3505	582	1,291
	3506	940	1,896
	3507	511	1,135
	3508	1,200	2,592
	3509	1,818	3,760
	3510	1,198	2,614
	3511	1,734	3,413
	3512	563	1,357
	3513	1,815	3,930
	3514	728	1,625

Appendix E (2017) - Response rates by health region and frame and response rates by Local Health Integrated Network (LHIN) and frame in Ontario

Appendix E (2017) - Table 9.1 response rates by health region and frame

		Adults			Children			Combined
Prov. Terr.	Health Region	# in scope pers.	# resp.	Resp. rates	# in scope pers.	# resp. pers.	Resp. rates	Combined resp. rates
Canada	**Total**	84,380	53,562	63.5	8,164	4,573	56.0	62.8
N.L.	**Total**	2,219	1,504	67.8	254	160	63.0	67.3
	1011	864	559	64.7	97	61	62.9	64.5
	1012	538	376	69.9	67	41	61.2	68.9
	1013	442	318	71.9	56	37	66.1	71.3
	1014	375	251	66.9	34	21	61.8	66.5
P.E.I.	**Total**	1,388	891	64.2	180	92	51.1	62.7
	1100	1,388	891	64.2	180	92	51.1	62.7
N.S.	**Total**	3,291	2,220	67.5	317	172	54.3	66.3
	1201	704	512	72.7	84	50	59.5	71.3
	1202	681	469	68.9	71	35	49.3	67.0
	1203	802	529	66.0	69	29	42.0	64.1
	1204	1,104	710	64.3	93	58	62.4	64.2
N.B.	**Total**	2,618	1,656	63.3	302	162	53.6	62.3
	1301	498	312	62.7	57	32	56.1	62.0
	1302	460	296	64.3	52	33	63.5	64.3
	1303	399	267	66.9	35	22	62.9	66.6
	1304	339	199	58.7	39	21	53.8	58.2
	1305	234	140	59.8	26	15	57.7	59.6
	1306	411	259	63.0	58	22	37.9	59.9
	1307	277	183	66.1	35	17	48.6	64.1
Que.	**Total**	17,414	11,629	66.8	1,331	907	68.1	66.9
	2401	786	566	72.0	61	46	75.4	72.3
	2402	977	702	71.9	65	46	70.8	71.8
	2403	1,244	881	70.8	102	77	75.5	71.2
	2404	996	684	68.7	91	64	70.3	68.8

	2405	969	678	70.0	81	57	70.4	70.0
	2406	2,432	1,535	63.1	173	107	61.8	63.0
	2407	1,096	700	63.9	70	46	65.7	64.0
	2408	860	574	66.7	61	42	68.9	66.9
	2409	811	504	62.1	50	31	62.0	62.1
	2410	321	198	61.7	19	13	68.4	62.1
	2411	788	526	66.8	57	37	64.9	66.6
	2412	970	702	72.4	77	60	77.9	72.8
	2413	1,182	725	61.3	105	67	63.8	61.5
	2414	1,095	743	67.9	85	54	63.5	67.5
	2415	1,177	791	67.2	102	71	69.6	67.4
	2416	1,710	1,120	65.5	132	89	67.4	65.6
Ont.	**Total**	26,241	16,230	61.8	2,441	1,310	53.7	61.2
	3526	643	412	64.1	45	25	55.6	63.5
	3527	508	297	58.5	66	35	53.0	57.8
	3530	1,084	637	58.8	99	56	56.6	58.6
	3531	447	312	69.8	57	35	61.4	68.8
	3533	530	376	70.9	63	40	63.5	70.2
	3534	509	335	65.8	50	24	48.0	64.2
	3535	580	362	62.4	50	24	48.0	61.3
	3536	952	558	58.6	92	45	48.9	57.8
	3537	1,066	614	57.6	92	44	47.8	56.8
	3538	578	403	69.7	54	29	53.7	68.4
	3539	366	254	69.4	39	24	61.5	68.6
	3540	471	311	66.0	47	24	51.1	64.7
	3541	677	438	64.7	47	29	61.7	64.5
	3542	536	366	68.3	44	19	43.2	66.4
	3543	533	345	64.7	65	35	53.8	63.5
	3544	871	512	58.8	106	59	55.7	58.4
	3546	954	577	60.5	74	41	55.4	60.1
	3547	476	323	67.9	42	27	64.3	67.6
	3549	509	312	61.3	35	12	34.3	59.6

	3551	1,127	700	62.1	114	75	65.8	62.4
	3552	440	271	61.6	52	28	53.8	60.8
	3553	1,479	834	56.4	128	62	48.4	55.8
	3554	395	274	69.4	39	20	51.3	67.7
	3555	554	341	61.6	49	24	49.0	60.5
	3556	430	274	63.7	42	21	50.0	62.5
	3557	512	326	63.7	43	29	67.4	64.0
	3558	643	437	68.0	55	34	61.8	67.5
	3560	988	591	59.8	80	44	55.0	59.5
	3561	619	404	65.3	61	27	44.3	63.4
	3562	624	400	64.1	53	30	56.6	63.5
	3563	290	200	69.0	22	11	50.0	67.6
	3565	959	594	61.9	101	66	65.3	62.3
	3566	766	507	66.2	57	33	57.9	65.6
	3568	886	533	60.2	104	45	43.3	58.4
	3570	1,347	741	55.0	120	56	46.7	54.3
	3595	1,892	1,059	56.0	154	78	50.6	55.6
Man.	**Total**	3,855	2,506	65.0	391	237	60.6	64.6
	4601	1,355	837	61.8	113	71	62.8	61.9
	4602	659	439	66.6	82	51	62.2	66.1
	4603	658	442	67.2	64	41	64.1	66.9
	4604	446	295	66.1	52	26	50.0	64.5
	4605	737	493	66.9	80	48	60.0	66.2
Sask.	**Total**	3,277	2,216	67.6	370	195	52.7	66.1
	4701	249	191	76.7	24	12	50.0	74.4
	4702	254	174	68.5	36	15	41.7	65.2
	4703	254	185	72.8	23	15	65.2	72.2
	4704	463	298	64.4	48	27	56.3	63.6
	4705	294	208	70.7	27	18	66.7	70.4
	4706	557	383	68.8	52	27	51.9	67.3
	4707	199	144	72.4	16	9	56.3	71.2
	4708	238	145	60.9	27	14	51.9	60.0

	4709	304	191	62.8	53	24	45.3	60.2
	4710	277	183	66.1	34	22	64.7	65.9
	4714	188	114	60.6	30	12	40.0	57.8
Alta.	**Total**	10,532	6,397	60.7	954	540	56.6	60.4
	4831	1,465	926	63.2	135	85	63.0	63.2
	4832	2,918	1,801	61.7	223	133	59.6	61.6
	4833	1,741	1,104	63.4	171	83	48.5	62.1
	4834	2,685	1,579	58.8	239	142	59.4	58.9
	4835	1,723	987	57.3	186	97	52.2	56.8
B.C.	**Total**	12,021	7,291	60.7	1,196	620	51.8	59.9
	5911	415	262	63.1	46	21	45.7	61.4
	5912	460	291	63.3	58	34	58.6	62.7
	5913	877	556	63.4	83	46	55.4	62.7
	5914	779	493	63.3	57	31	54.4	62.7
	5921	841	507	60.3	81	37	45.7	59.0
	5922	1,110	646	58.2	125	66	52.8	57.7
	5923	1,138	676	59.4	123	62	50.4	58.5
	5931	716	413	57.7	70	35	50.0	57.0
	5932	1,221	688	56.3	92	46	50.0	55.9
	5933	768	475	61.8	84	45	53.6	61.0
	5941	793	468	59.0	95	55	57.9	58.9
	5942	765	488	63.8	67	32	47.8	62.5
	5943	572	360	62.9	48	23	47.9	61.8
	5951	491	290	59.1	39	17	43.6	57.9
	5952	642	399	62.1	64	34	53.1	61.3
	5953	433	279	64.4	64	36	56.3	63.4
Y.T.	**6001**	628	383	61.0	121	67	55.4	60.1
N.W.T.	**6101**	578	386	66.8	141	59	41.8	61.9
Nvt.	**6201**	318	253	79.6	166	52	31.3	63.0

Appendix E (2017) - Table 9.2 Response rate by Local Health Integrated Network (LHIN) in Ontario

		Adults		
Prov. Terr.	LHIN	# in scope pers.	# resp.	Resp. rates
Ont.	**Total**	26,241	16,230	61.8
	3501	1,893	1,210	63.9
	3502	3,049	1,999	65.6
	3503	1,551	1,004	64.7
	3504	3,396	2,033	59.9
	3505	971	520	53.6
	3506	1,448	857	59.2
	3507	817	465	56.9
	3508	1,945	1,052	54.1
	3509	2,658	1,590	59.8
	3510	1,614	1,063	65.9
	3511	2,456	1,586	64.6
	3512	852	526	61.7
	3513	2,458	1,613	65.6
	3514	1,133	712	62.8